

Lecture 11
Quantum Information for Newcomers
Qubits, Registers, Density Matrices and Noise

Deep Learning in Classical and Quantum Physics

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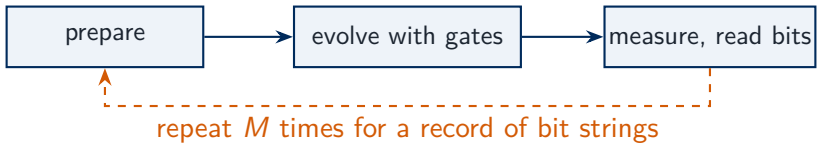
Prepare, Evolve, Measure



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Physical Systems as Carriers of Information

A classical bit is a two-state system that we prepare, transform and read out. A qubit is its quantum counterpart.



One run returns one bit per qubit. Theory predicts probabilities, which we compare with records of many repeated runs, called *shots*.

Running example, the two-spin register

Qubits 0 and 1 are prepared, interact and are measured. We describe the register in a computer and predict its records.

Prerequisites and a Dictionary

We assume complex numbers, vectors and matrices, eigenvalues, basic probability and a little Python, and no prior course in quantum mechanics.

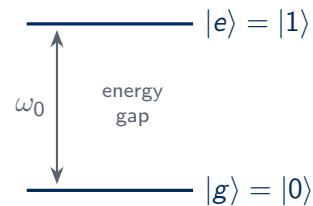
physics	mathematics	computer science
state, ket $\|\psi\rangle$	unit column vector in $\mathbb{C}^{2^N}$	array of 2^N complex numbers
bra $\langle\psi\|$, $\langle\phi\|\psi\rangle$	conjugate row, inner product	`psi.conj()`, `vdot`
observable	Hermitian matrix	quantity averaged over shots
Hamiltonian $\hat{\mathcal{H}}$ (energy)	Hermitian generator of time evolution	cost function of variational algorithms
gate, time evolution	unitary matrix	reversible instruction
measurement in a basis	projectors, Born rule	sampling a bit string
two systems	tensor (Kronecker) product	`np.kron`, binary indexing
density matrix	positive matrix of trace one	probability-weighted states
partial trace	linear map $\mathbb{C}^{4\times4} \to \mathbb{C}^{2\times2}$	marginalising a qubit

The Qubit as a Two-Level System

A **qubit** is any quantum system with two perfectly distinguishable states. Its states form a two-dimensional complex vector space. Before a measurement the qubit can be in a superposition of both levels, and a measurement in that basis finds one of them.

Our running example is an atom with two energy levels, the lower *ground* state and the higher *excited* state, separated by the energy $\hbar\omega_0$. We set $\hbar = 1$, so energies are frequencies, and we label

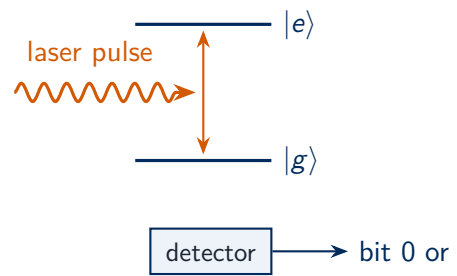
$$|0\rangle = |\uparrow\rangle \equiv |g\rangle, \qquad |1\rangle = |\downarrow\rangle \equiv |e\rangle.$$



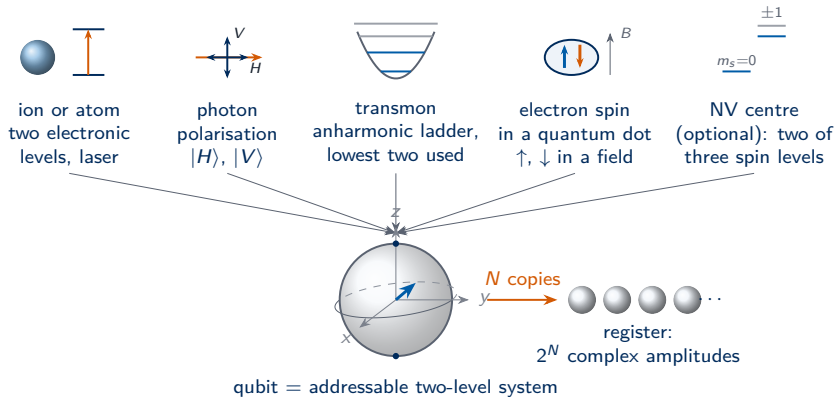
Driving and Measuring an Atom

A laser tuned to the energy gap drives the atom between $|g\rangle$ and $|e\rangle$. The pulse duration sets how far the state rotates, and a short pulse leaves the atom in a superposition of both levels.

A second laser makes the atom fluoresce in only one of the two levels. A detector records light or no light, one bit per run.

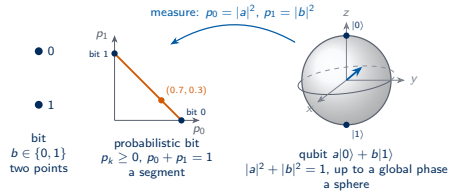


Platforms for Qubits



Ions, neutral atoms, photons, superconducting circuits and spins in semiconductors all provide addressable two-level systems. A register of N qubits consists of N such systems that can be driven and measured individually and made to interact.

Bit, Probabilistic Bit and Qubit



	bit	probabilistic bit	qubit
state	0 or 1	(p_0, p_1), $p_k \geq 0$, $\sum p_k = 1$	$(a, b) \in \mathbb{C}^2$, $\lvert a\rvert^2 + \lvert b\rvert^2 = 1$
operations	logic gates	stochastic matrices	unitary matrices (reversible)
readout	the bit	sample a bit with p_k	sample a bit with $\lvert a\rvert^2, \lvert b\rvert^2$
N units	N bits	2^N probabilities	2^N complex amplitudes

A qubit still outputs a bit. Its complex amplitudes can cancel each other in interference, and a classical description of N qubits stores 2^N complex amplitudes.

Four Rules of Quantum Mechanics

The description of N qubits rests on four postulates motivated by experiments.

1. The **state** is a unit vector $|\psi\rangle \in \mathbb{C}^{2^N}$, defined up to a global phase.
2. An isolated system **evolves** by a unitary matrix, $|\psi\rangle \to \hat{U}|\psi\rangle$, and $\hat{U} = e^{-it\hat{\mathcal{H}}}$ with the energy matrix $\hat{\mathcal{H}}$ of the system.
3. A **measurement** in an orthonormal basis $\{|k\rangle\}$ gives the outcome k with probability $|\langle k|\psi\rangle|^2$ (the Born rule) and leaves the state $|k\rangle$.
4. Two systems **together** live in the tensor product $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$.

Density matrices and noise extend the first three rules to systems that we know only statistically or only in part.

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One Qubit



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The State of One Qubit

The two basis states, spin up and spin down along z, form the **computational basis**,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A general state is a **superposition** with complex *amplitudes* a and b,

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle = \begin{pmatrix} a \\ b \end{pmatrix}, \qquad \underbrace{|a|^2 + |b|^2 = 1}_{\text{normalisation}}.$$

In a computer one qubit is an array of two complex numbers.

Running example

$$|\psi_{\text{ex}}\rangle = \frac{\sqrt{3}}{2}\,|0\rangle + \frac{e^{i\pi/4}}{2}\,|1\rangle, \qquad \tfrac{3}{4} + \tfrac{1}{4} = 1.$$

Dirac Notation

symbol	object	example
ket $\lvert\psi\rangle$	column vector	$\lvert\psi\rangle = (a, b)^\top$
bra $\langle\psi\rvert = (\lvert\psi\rangle)^\dagger$	row vector, conjugated	$\langle\psi\rvert = (a^*, b^*)$
$\langle\phi\rvert\psi\rangle$	inner product, a number	$\sum_k \phi_k^* \psi_k$, $\langle 0\rvert 1\rangle = 0$
$\lvert a\rangle\langle b\rvert$	outer product, a matrix	$\lvert 0\rangle\langle 0\rvert = \mathrm{diag}(1, 0)$

- The dagger $\dagger$ is complex conjugation followed by transposition.
- $\lvert 0\rangle\langle 0\rvert$ and $\lvert 1\rangle\langle 1\rvert$ are **projectors** with $\hat{P}^2 = \hat{P}$, and $\lvert 0\rangle\langle 0\rvert\psi\rangle = a\lvert 0\rangle$ keeps one component.
- A matrix with $\hat{O}^\dagger = \hat{O}$ is **Hermitian**, and its eigenvalues are real.

The Born Rule

A measurement in the computational basis returns one bit, with the probabilities

$$p(0) = |\langle 0|\psi\rangle|^2 = |a|^2, \qquad p(1) = |\langle 1|\psi\rangle|^2 = |b|^2, \qquad p(0) + p(1) = 1.$$

The measurement leaves the qubit in the basis state it found, and repeating the measurement gives that bit again.

Running example

$p(0) = |\sqrt{3}/2|^2 = 0.750$ and $p(1) = |e^{i\pi/4}/2|^2 = 1/4$. The phase $e^{i\pi/4}$ does not enter these probabilities.

Global and Relative Phase

Multiplying the whole state by a **global phase**, $|\psi\rangle \to e^{i\alpha}|\psi\rangle$, changes no prediction, because every prediction contains amplitudes in pairs,

$$(e^{i\alpha}a)^*(e^{i\alpha}b) = \underbrace{e^{-i\alpha}e^{i\alpha}}_{=1}a^*b.$$

The global phase can be chosen to make a real and non-negative.

A **relative phase** between the two amplitudes is physical. The two states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

have $|a|^2 = |b|^2 = \frac{1}{2}$. Both give a fair coin in the computational basis, although they are orthogonal, $\langle +|-\rangle = 0$.

Which measurement tells $|+\rangle$ from $|-\rangle$ with certainty?

Pauli Matrices and Expectation Values

Recording $+1$ for bit 0 and -1 for bit 1 measures the spin along z, whose mean is

$$\langle \hat{Z} \rangle = (+1)\, p(0) + (-1)\, p(1) = p(0) - p(1).$$

The spin components along x, y, z are the Pauli matrices, $\hat{\sigma}_x \equiv \hat{X}$, $\hat{\sigma}_y \equiv \hat{Y}$, $\hat{\sigma}_z \equiv \hat{Z}$,

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

$\hat{X}^2 = \hat{Y}^2 = \hat{Z}^2 = \hat{I}$, with eigenvalues ± 1 and eigenvectors $|0\rangle, |1\rangle$ for $\hat{Z}$, $|\pm\rangle$ for $\hat{X}$ and $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$ for $\hat{Y}$.

The **expectation value** of an observable $\hat{O}$ with eigenvalues λ_k is $\langle \hat{O} \rangle = \langle \psi | \hat{O} | \psi \rangle = \sum_k \lambda_k\, p(\lambda_k)$. For the running example $\langle \hat{Z} \rangle = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$.

Pauli Algebra

Multiplying the matrices out gives one rule for all products, with $a, b, c \in \{x, y, z\}$ and ε_{abc} the sign of the permutation ($\varepsilon_{xyz} = 1$),

$$\hat{\sigma}_a \hat{\sigma}_b = \delta_{ab} \hat{I} + i\, \varepsilon_{abc}\, \hat{\sigma}_c \qquad \Longrightarrow \qquad \hat{\sigma}_a^2 = \hat{I}, \quad \hat{X}\hat{Y} = i\hat{Z}, \quad \hat{Y}\hat{Z} = i\hat{X}, \quad \hat{Z}\hat{X} = i\hat{Y}.$$

- The anticommutator $\{\hat{\sigma}_a, \hat{\sigma}_b\} = 2\delta_{ab}\hat{I}$ makes different Pauli matrices *anticommute*, $\hat{X}\hat{Y} = -\hat{Y}\hat{X}$.
- The commutator is $[\hat{\sigma}_a, \hat{\sigma}_b] = 2i\, \varepsilon_{abc}\, \hat{\sigma}_c$.

The spin operators of a spin-1/2 are $\hat{S}_a = \hat{\sigma}_a/2$.

- $\mathrm{Tr}\hat{\sigma}_a = 0$ and $\mathrm{Tr}(\hat{\sigma}_a\hat{\sigma}_b) = 2\delta_{ab}$ make $\hat{I}, \hat{X}, \hat{Y}, \hat{Z}$ an orthogonal basis.
- Any 2×2 matrix is $c_0\hat{I} + \mathbf{c} \cdot \hat{\boldsymbol{\sigma}}$, with $c_0 = \frac{1}{2}\mathrm{Tr}\hat{A}$ and $c_a = \frac{1}{2}\mathrm{Tr}(\hat{\sigma}_a\hat{A})$.

Ladder Operators

The two unit transition matrices move one basis column into the other,

$$\hat{S}^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = |0\rangle\langle 1|, \qquad \hat{S}^+ \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \hat{S}^+ \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0,$$

and $\hat{S}^- = (\hat{S}^+)^\dagger = |1\rangle\langle 0|$ does the reverse. $\hat{S}^+$ turns $|\downarrow\rangle$ into $|\uparrow\rangle$ and raises the spin. They are $\hat{S}^\pm = \hat{S}_x \pm i\hat{S}_y$, and in terms of the Pauli matrices

$$\hat{\sigma}^\pm = \hat{X} \pm i\hat{Y} = 2\hat{S}^\pm, \qquad \hat{\sigma}^+ = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \qquad \hat{\sigma}^- = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix}.$$

- $\hat{X} = \frac{1}{2}(\hat{\sigma}^+ + \hat{\sigma}^-)$ and $\hat{Y} = \frac{1}{2i}(\hat{\sigma}^+ - \hat{\sigma}^-)$.
- $(\hat{\sigma}^\pm)^2 = 0$, so the spin cannot be raised twice.
- $\hat{\sigma}^+\hat{\sigma}^- = 2(\hat{I} + \hat{Z})$ and $\hat{\sigma}^-\hat{\sigma}^+ = 2(\hat{I} - \hat{Z})$.
- $[\hat{\sigma}^+, \hat{\sigma}^-] = 4\hat{Z}$ and $[\hat{Z}, \hat{\sigma}^\pm] = \pm 2\hat{\sigma}^\pm$.

Ladder Operators and Energy Jumps

In this course spin up is the ground state, $|0\rangle = |\uparrow\rangle = |g\rangle$, and raising the spin lowers the energy.

operator	matrix	on the spin	on the energy
$\hat{S}^+ = \frac{1}{2}\hat{\sigma}^+ = \lvert g\rangle\langle e\rvert$	$\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$	raises, $\lvert\downarrow\rangle \to \lvert\uparrow\rangle$	decay, $\lvert e\rangle \to \lvert g\rangle$
$\hat{S}^- = \frac{1}{2}\hat{\sigma}^- = \lvert e\rangle\langle g\rvert$	$\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$	lowers, $\lvert\uparrow\rangle \to \lvert\downarrow\rangle$	excitation, $\lvert g\rangle \to \lvert e\rangle$

Two conventions in the literature

Atomic-physics texts often write $\hat{\sigma}_z^{\text{atom}} = |e\rangle\langle e| - |g\rangle\langle g| = -\hat{Z}$. There $\hat{\sigma}^+_{\text{atom}} = |e\rangle\langle g|$ excites the atom and $\hat{\sigma}^-_{\text{atom}} = |g\rangle\langle e|$ is the decay. These $\hat{\sigma}^{\pm}_{\text{atom}}$ correspond to our $\hat{S}^{\mp}$. The level that a paper calls up has to be checked before its formulas are used.

The Bloch Sphere

Up to a global phase,

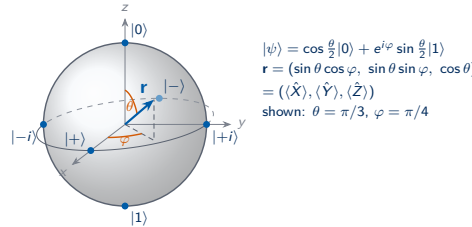
$$|\psi\rangle = \cos\tfrac{\vartheta}{2}\,|0\rangle + e^{i\varphi}\sin\tfrac{\vartheta}{2}\,|1\rangle.$$

The **Bloch vector** collects the three expectation values,

$$\mathbf{r} = (\langle\hat{X}\rangle, \langle\hat{Y}\rangle, \langle\hat{Z}\rangle)$$

$$= (\sin\vartheta\cos\varphi,\ \sin\vartheta\sin\varphi,\ \cos\vartheta).$$

The poles are $|0\rangle$ and $|1\rangle$, and the equator holds $|\pm\rangle$ and $|\pm i\rangle$. The relative phase is the azimuth φ, visible in $\langle\hat{X}\rangle$ and $\langle\hat{Y}\rangle$, and pure states have $|\mathbf{r}| = 1$.



The running example has $\vartheta = \pi/3$, $\varphi = \pi/4$ and $\mathbf{r}_{\mathrm{ex}} = (0.612,\ 0.612,\ 0.500)$.

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Evolution and Gates as Rotations



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The Evolution Operator

The *Hamiltonian* $\hat{\mathcal{H}} = \hat{\mathcal{H}}^\dagger$ is the energy matrix, whose eigenvalues are the energy levels. The state obeys the Schrödinger equation ($\hbar = 1$)

$$i\frac{d}{dt}|\psi(t)\rangle = \hat{\mathcal{H}}|\psi(t)\rangle.$$

For a **time-independent** $\hat{\mathcal{H}}$ we define $\hat{U}(t) = e^{-it\hat{\mathcal{H}}} = \sum_{k\geq 0}(-it\hat{\mathcal{H}})^k/k!$ by its power series and differentiate each term,

$$\frac{d}{dt}\hat{U}(t) = \sum_{k\geq 1}\frac{k\,t^{k-1}(-i\hat{\mathcal{H}})^k}{k!} = (-i\hat{\mathcal{H}})\underbrace{\sum_{k\geq 1}\frac{(-it\hat{\mathcal{H}})^{k-1}}{(k-1)!}}_{=\hat{U}(t)} = -i\hat{\mathcal{H}}\,\hat{U}(t), \qquad \underbrace{\hat{U}(0) = \hat{I}}_{\text{start}},$$

so $|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle$ solves the equation with the right initial state.

Properties of the Evolution Operator

- $\hat{U}^\dagger = e^{+it\hat{\mathcal{H}}}$ because $\hat{\mathcal{H}}^\dagger = \hat{\mathcal{H}}$. The two exponentials commute, and $\hat{U}^\dagger \hat{U} = \hat{I}$ preserves the total probability.
- Evolving for $t_1 + t_2$ is evolving twice, $\hat{U}(t_1 + t_2) = \hat{U}(t_1)\,\hat{U}(t_2)$.

With $\hat{\mathcal{H}}|E_k\rangle = E_k|E_k\rangle$, every power is $\hat{\mathcal{H}}^n = \sum_k E_k^n |E_k\rangle\langle E_k|$, and the series sums to the spectral form

$$\hat{U}(t) = \sum_k e^{-iE_k t}\,|E_k\rangle\langle E_k|.$$

An energy eigenstate acquires only a phase, and none of its predictions changes in time.

Precession, $\hat{\mathcal{H}} = \frac{\Delta}{2}\hat{Z}$

The eigenvalues are $\pm\frac{\Delta}{2}$ on $|0\rangle$ and $|1\rangle$, so $\hat{U}(t) = \mathrm{diag}(e^{-i\Delta t/2}, e^{i\Delta t/2})$. The relative phase $e^{+i\Delta t}$ on $|1\rangle$ turns the Bloch vector about z, $\varphi \to \varphi + \Delta t$.

The Atom and Its Hamiltonian

The qubit Hamiltonian of an atom derives from the Hamiltonian of a nucleus of charge Ze_0 and mass M with N electrons of charge $-e_0$ and mass m_e,

$$\hat{\mathcal{H}}_{\text{atom}} = \underbrace{\frac{\hat{\mathbf{P}}^2}{2M} + \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e}}_{\text{kinetic energies}} \underbrace{- \sum_i \frac{Ze_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{R}}|}}_{\text{electron–nucleus attraction}} + \underbrace{\sum_{i<j} \frac{e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|}}_{\text{electron–electron repulsion}} .$$

- $\hat{\mathbf{R}}, \hat{\mathbf{P}}$ are the position and momentum of the nucleus, $\hat{\mathbf{r}}_i, \hat{\mathbf{p}}_i$ those of electron i, and ε_0 is the vacuum permittivity.
- Relativistic corrections (fine structure) and the nuclear spin (hyperfine structure) are omitted at this level.

C. J. Foot, *Atomic Physics* (Oxford University Press, 2005)

Born–Oppenheimer Approximation

The nucleus is $M/m_e \approx 1840\,A$ times heavier (A the mass number) and moves much more slowly than the electrons. We clamp it at $\mathbf{R} = 0$ and solve the **electronic Hamiltonian**

$$\hat{\mathcal{H}}_{\mathrm{el}} = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} - \sum_i \frac{Ze_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i|} + \sum_{i<j} \frac{e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|},$$

whose eigenvalues are the energy levels of the atom.

- For hydrogen the centre of mass separates, and only the reduced mass $\mu = m_e M/(m_e + M)$ remains, a 0.05 % shift. Several electrons add a small mass-polarisation term.
- The same approximation underlies molecular physics and quantum chemistry.

M. Born and R. Oppenheimer, Ann. Phys. **389**, 457 (1927), doi:10.1002/andp.19273892002

Energy Eigenstates as a Basis

The bound states $\hat{\mathcal{H}}_{\rm el}|n\rangle = E_n|n\rangle$ are discrete and orthonormal, with n collecting all quantum numbers. Every state is a superposition,

$$|\psi(t)\rangle = \sum_n c_n(t)\,|n\rangle, \qquad c_n = \langle n|\psi\rangle, \qquad \sum_n |c_n|^2 = 1,$$

and $|c_n|^2$ is the probability of finding the atom in $|n\rangle$. Projecting the Schrödinger equation on $\langle n|$ gives

$$i\hbar\,\dot{c}_n = \sum_m H_{nm}\,c_m, \qquad H_{nm} = \langle n|\hat{\mathcal{H}}|m\rangle,$$

the matrix form of the Schrödinger equation, with $c_n(t) = c_n(0)e^{-iE_nt/\hbar}$ for the free atom. Alkali atoms and ions such as Ca^+, Sr^+, Yb^+ have one valence electron and such discrete levels.

Coupling to Laser Light

A classical field $\mathbf{E}(t) = E_0 \mathbf{e}_z \cos(\omega_L t)$ is uniform over the atom, since an optical wavelength exceeds the atom by orders of magnitude (dipole approximation). It adds

$$\hat{\mathcal{H}}_{\mathrm{int}} = -\hat{\mathbf{d}} \cdot \mathbf{E}(t), \qquad \hat{\mathbf{d}} = -e_0 \sum_i \hat{\mathbf{r}}_i, \qquad d_{nm} = \langle n|\hat{\mathbf{d}} \cdot \mathbf{e}_z|m\rangle.$$

$\hat{\mathcal{H}}_{\mathrm{el}}$ is unchanged when all $\mathbf{r}_i \to -\mathbf{r}_i$, so its eigenstates can be chosen even or odd, while $\hat{\mathbf{d}}$ changes sign. Hence $d_{nn} = 0$, and states of equal parity are not coupled, a **selection rule**.

Two Levels Near Resonance

Removing the free rotation, $c_n = a_n e^{-iE_n t/\hbar}$, leaves only the effect of the light,

$$i\hbar\,\dot{a}_n = -E_0\cos(\omega_L t)\sum_m d_{nm}\,e^{i\omega_{nm}t}a_m, \qquad \omega_{nm} = \frac{E_n - E_m}{\hbar}.$$

For a laser near the transition $g \leftrightarrow e$, $\omega_L \approx \omega_0 = \omega_{eg}$,

$$e^{i\omega_0 t}\cos(\omega_L t) = \underbrace{\tfrac{1}{2}e^{-i\Delta t}}_{\text{slow, } \Delta=\omega_L-\omega_0} + \underbrace{\tfrac{1}{2}e^{i(\omega_0+\omega_L)t}}_{\text{fast, averages out}}.$$

Dropping fast terms and far-detuned levels (rotating-wave approximation) leaves two amplitudes,

$$i\dot{a}_g = \tfrac{\Omega}{2}e^{i\Delta t}a_e, \qquad i\dot{a}_e = \tfrac{\Omega}{2}e^{-i\Delta t}a_g, \qquad \Omega = -\frac{d_{eg}E_0}{\hbar}\ \text{(Rabi frequency)},$$

with the phases of $|g\rangle$, $|e\rangle$ chosen so that d_{eg} is real.

The Qubit Hamiltonian

With $b_g = a_g e^{-i\Delta t/2}$ and $b_e = a_e e^{i\Delta t/2}$ the time dependence disappears,

$$i\frac{d}{dt}\begin{pmatrix} b_g \\ b_e \end{pmatrix} = \begin{pmatrix} \Delta/2 & \Omega/2 \\ \Omega/2 & -\Delta/2 \end{pmatrix}\begin{pmatrix} b_g \\ b_e \end{pmatrix}, \qquad \hat{\mathcal{H}}_{\mathrm{RF}} = \frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}\hat{X}, \qquad \Delta = \omega_L - \omega_0,$$

with $\hbar = 1$ and the qubit $|g\rangle = |0\rangle$, $|e\rangle = |1\rangle$.

- In the laboratory frame the two-level atom has $\hat{\mathcal{H}}(t) = -\frac{\omega_0}{2}\hat{Z} + \Omega\cos(\omega_L t)\,\hat{X}$, and $-\frac{\omega_0}{2}\hat{Z}$ puts $|g\rangle$ lower.
- The amplitudes $b = e^{-i\omega_L t\hat{Z}/2}c$ belong to the frame rotating with the laser. Probabilities are equal in both frames.

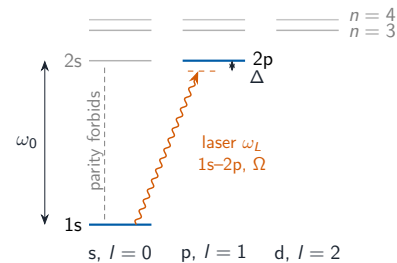
L. Allen and J. H. Eberly, *Optical Resonance and Two-Level Atoms* (Dover, 1987)

Hydrogen as an Example

Hydrogen has $Z = N = 1$, eigenstates $|nlm\rangle$ with $E_n = -\mathrm{Ry}/n^2$, $\mathrm{Ry} \approx 13.6$ eV, and $\langle \mathbf{r}|nlm\rangle = R_{nl}(r)Y_{lm}(\vartheta, \varphi)$, a radial function times a spherical harmonic of the two angles.

- Light along z couples only $l' = l \pm 1$, $m' = m$, and leaves 1s and 2s uncoupled.
- $\langle 2p_0|\hat{z}|1s\rangle = \frac{128\sqrt{2}}{243} a_0 \approx 0.745\, a_0$, and $\Omega = e_0 E_0 \langle 2p_0|\hat{z}|1s\rangle/\hbar$.
- $\hbar\omega_0 = \frac{3}{4}\mathrm{Ry} \approx 10.2$ eV, the first ultraviolet line of hydrogen (Lyman-α), at 121.5 nm.

Qubit $|g\rangle = |1s\rangle = |0\rangle$, $|e\rangle = |2p_0\rangle = |1\rangle$.



Limits of the Two-Level Description

- A far level r with Rabi frequency Ω_r and detuning $\Delta_r = \omega_L - \omega_{rg}$ shifts the ground level by $\delta E_g = \Omega_r^2/(4\Delta_r)$, the **light shift** (a quoted second-order result), and moves the resonance.
- The rotating-wave approximation needs $|\Omega|, |\Delta| \ll \omega_L$. The dropped terms raise the resonance by $\Omega^2/(4\omega_0)$, the Bloch–Siegert shift.
- The excited state decays by **spontaneous emission** with the rate $\Gamma = \omega_0^3 |d_{eg}|^2/(3\pi\varepsilon_0\hbar c^3)$. For hydrogen 2p, $1/\Gamma \approx 1.60$ ns, far too short for a computation. This decay sets the relaxation time T_1 of the qubit.

Atomic and ionic qubits therefore use long-lived levels, hyperfine ground-state levels or metastable states without a dipole decay. Microwaves, laser pairs or narrow optical transitions drive them with the Hamiltonian $\hat{\mathcal{H}}_{\mathrm{RF}}$.

Rabi Oscillations

On resonance, $\Delta = 0$, the Hamiltonian $\frac{\Omega}{2}\hat{X}$ has the eigenvectors $|\pm\rangle$ with energies $\pm\frac{\Omega}{2}$, and the spectral form gives

$$\hat{U}(t) = e^{-i\Omega t/2}|+\rangle\langle+| + e^{i\Omega t/2}|-\rangle\langle-| \equiv \hat{R}_x(\Omega t).$$

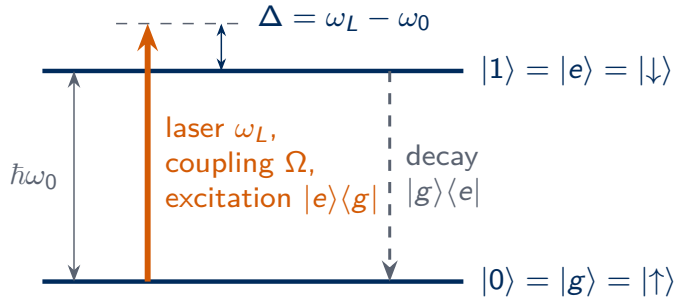
An atom that starts in $|g\rangle$ is found in $|e\rangle$ with the probability

$$P_e(t) = \sin^2\frac{\Omega t}{2},$$

which oscillates in time, the **Rabi oscillation**. A π pulse, $\Omega t = \pi$, takes $|g\rangle$ to $|e\rangle$, and a $\pi/2$ pulse creates an equal superposition.

I. I. Rabi, Phys. Rev. **51**, 652 (1937), doi:10.1103/PhysRev.51.652

A Driven Two-Level Atom



in the frame rotating with the laser ($\hbar = 1$):
$\hat{\mathcal{H}} = \frac{\Omega}{2}\hat{X} + \frac{\Delta}{2}\hat{Z}$; on resonance, $\Delta = 0$,
the drive gives $\hat{R}_x(\Omega t)$ (Rabi oscillation)

Detuning and Laser Phase

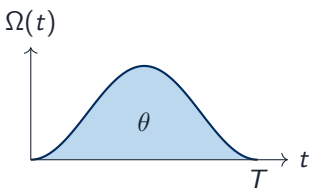
- Off resonance the oscillation is faster, at $\sqrt{\Omega^2 + \Delta^2}$, and incomplete, $P_{e,\max} = \Omega^2/(\Omega^2 + \Delta^2)$.
- The laser phase ϕ_L, in the field $E_0 \cos(\omega_L t - \phi_L)$, replaces $\hat{X}$ by $\cos\phi_L\,\hat{X} + \sin\phi_L\,\hat{Y}$ and selects the rotation axis in the x–y plane.
- A detuning alone, $\Omega = 0$, gives the precession $\hat{R}_z(\Delta t)$ about z.

In this driven-atom model, pulses of chosen duration, phase and detuning implement rotations of the qubit.

Rotation Angle from Pulse Area

On resonance a laser with phase ϕ_L and a slowly varying amplitude gives $\hat{\mathcal{H}}(t) = \frac{\Omega(t)}{2}(\cos\phi_L \hat{X} + \sin\phi_L \hat{Y})$. It commutes with itself at all times, so

$$\hat{U} = \hat{R}_{\phi_L}(\theta) = e^{-i\theta(\cos\phi_L \hat{X} + \sin\phi_L \hat{Y})/2}, \qquad \theta = \underbrace{\int_0^T \Omega(t')\, dt'}_{\text{pulse area}}.$$



- The duration or the amplitude sets the angle, and the phase ϕ_L sets the axis.
- A rectangular pulse gives $\theta = \Omega T$. A smooth envelope of equal area gives the same gate.
- With a detuning $\hat{\mathcal{H}}$ contains $\hat{Z}$, and the shape matters too.

Phase Updates as $\hat{Z}$ Rotations

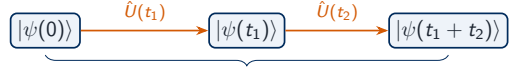
A rotation about z turns $\hat{X}$ into $\hat{R}_z(\phi)\hat{X}\hat{R}_z(-\phi) = \cos\phi\,\hat{X} + \sin\phi\,\hat{Y}$, so

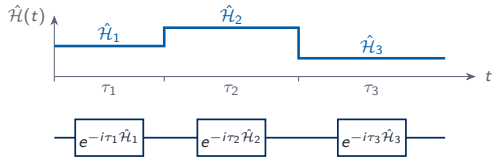
$$\hat{R}_\phi(\theta) = \hat{R}_z(\phi)\,\hat{R}_x(\theta)\,\hat{R}_z(-\phi) \quad \Longrightarrow \quad \hat{R}_\phi(\theta)\,\hat{R}_z(\alpha) = \hat{R}_z(\alpha)\,\hat{R}_{\phi-\alpha}(\theta).$$

- A rotation $\hat{R}_z(\alpha)$ followed by pulses with phases ϕ_k equals the same pulses with phases $\phi_k - \alpha$, then a final $\hat{R}_z(\alpha)$.
- The final z rotation leaves the probabilities of a readout in the computational basis unchanged.
- The control electronics apply $\hat{R}_z(\alpha)$ by shifting the phase of every later pulse, a **virtual $\hat{Z}$ gate** that takes no time.

From a Hamiltonian to a Circuit

$$i\tfrac{d}{dt}|\psi(t)\rangle = \hat{\mathcal{H}}|\psi(t)\rangle \;\Rightarrow\; |\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle, \quad \hat{U}(t) = e^{-it\hat{\mathcal{H}}} = \textstyle\sum_n \frac{(-it)^n}{n!}\hat{\mathcal{H}}^n$$





A quantum gate is a Hamiltonian switched on for a chosen time. Pulses with piecewise-constant Hamiltonians give a product of exponentials, $\hat{U} = e^{-it_3\hat{\mathcal{H}}_3}e^{-it_2\hat{\mathcal{H}}_2}e^{-it_1\hat{\mathcal{H}}_1}$, read as a circuit from right to left.

Exponential of a Pauli Matrix

Every Pauli matrix squares to the identity. For $\hat{P}^2 = \hat{I}$ we have $\hat{P}^{2m} = \hat{I}$ and $\hat{P}^{2m+1} = \hat{P}$, and the series splits into even and odd powers,

$$e^{-ia\hat{P}} = \underbrace{\sum_m \frac{(-ia)^{2m}}{(2m)!}}_{=\cos a}\,\hat{I} + \underbrace{\sum_m \frac{(-ia)^{2m+1}}{(2m+1)!}}_{=-i\sin a}\,\hat{P} = \cos a\,\hat{I} - i\sin a\,\hat{P}.$$

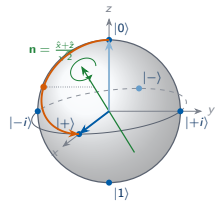
This matrix Euler formula holds for $\hat{X}$, $\hat{Y}$ and $\hat{Z}$, for $\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ with $|\mathbf{n}| = 1$, where anticommutation gives $(\mathbf{n}\cdot\hat{\boldsymbol{\sigma}})^2 = \hat{I}$, and for $\hat{X}\otimes\hat{X}$ and $\hat{Z}\otimes\hat{Z}$.

Rotations About Any Axis

The **rotation** about the unit axis $\mathbf{n}$ by the angle θ is

$$\begin{aligned}\hat{R}_{\mathbf{n}}(\theta) &= e^{-i\theta\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}/2} \\ &= \cos\frac{\theta}{2}\,\hat{I} - i\sin\frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}, \\ \hat{R}_z(\theta) &= \mathrm{diag}\big(e^{-i\theta/2}, e^{i\theta/2}\big).\end{aligned}$$

$\hat{R}_{\mathbf{n}}(\theta)$ turns the Bloch vector by θ about $\mathbf{n}$. The factor $\frac{1}{2}$ makes θ the geometric angle, and $\hat{R}_{\mathbf{n}}(2\pi) = -\hat{I}$ is a global phase.



$\hat{R}_{\mathbf{n}}(\theta) = e^{-i\theta\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}/2}$ turns $\mathbf{r}$ by θ about $\mathbf{n}$ (right-hand rule)

$\hat{H} = i\,\hat{R}_{\mathbf{n}}(\pi)$: $|0\rangle \to |+\rangle$ along the arc, which passes through $\mathbf{r} = (\frac{1}{2}, -\frac{1}{\sqrt{2}}, \frac{1}{2})$ at $\theta = \pi/2$

Single-Qubit Gates as Rotations

Every single-qubit gate is $\hat{G} = e^{i\alpha}\hat{R}_{\mathbf{n}}(\theta)$ with an axis $\mathbf{n}$, an angle θ and an unobservable global phase $e^{i\alpha}$. As an evolution for unit time, $\hat{G} = e^{-i\hat{\mathcal{H}}_G}$ with $\hat{\mathcal{H}}_G = \frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}} - \alpha\hat{I}$, and the gates below have $\alpha = \theta/2$.

gate	matrix	rotation	axis $\mathbf{n}$	angle θ	generator $\hat{\mathcal{H}}_G$
$\hat{X}$, bit flip	$\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$	$i\,\hat{R}_x(\pi)$	x	π	$\frac{\pi}{2}(\hat{X} - \hat{I})$
$\hat{Z}$, phase flip	$\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$	$i\,\hat{R}_z(\pi)$	z	π	$\frac{\pi}{2}(\hat{Z} - \hat{I})$
$\hat{H}$, Hadamard	$\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$	$i\,\hat{R}_{\mathbf{n}}(\pi)$	$(x+z)/\sqrt{2}$	π	$\frac{\pi}{2}(\hat{H} - \hat{I})$
$\hat{S}$, phase gate	$\left(\begin{smallmatrix} 1 & 0 \\ 0 & i \end{smallmatrix}\right)$	$e^{i\pi/4}\hat{R}_z(\pi/2)$	z	$\pi/2$	$\frac{\pi}{4}(\hat{Z} - \hat{I})$
$\hat{T}$, phase gate	$\left(\begin{smallmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{smallmatrix}\right)$	$e^{i\pi/8}\hat{R}_z(\pi/4)$	z	$\pi/4$	$\frac{\pi}{8}(\hat{Z} - \hat{I})$

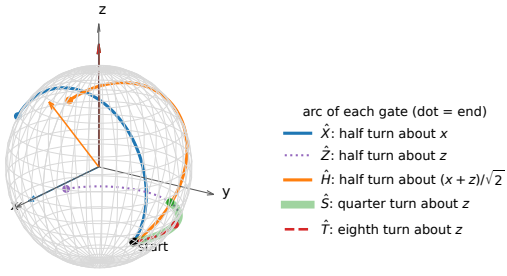
The phase gates obey $\hat{T}^2 = \hat{S}$ and $\hat{S}^2 = \hat{Z}$. The term $-\frac{\theta}{2}\hat{I}$ in $\hat{\mathcal{H}}_G$ produces only the global phase $e^{i\theta/2}$.

Single-Qubit Gates on Basis States and in the Laboratory

gate	action on basis states	Bloch sphere	implementation
$\hat{X}$	$\|0\rangle \leftrightarrow \|1\rangle$, $\|\pm\rangle \to \pm\|\pm\rangle$	half turn about x	pulse of area π, $\phi_L = 0$
$\hat{Z}$	$\|1\rangle \to -\|1\rangle$, $\|+\rangle \leftrightarrow \|-\rangle$	half turn about z	virtual $\hat{Z}$
$\hat{H}$	$\|0\rangle \to \|+\rangle$, $\|1\rangle \to \|-\rangle$	half turn about $(x+z)/\sqrt{2}$	virtual $\hat{Z}$, then area $\pi/2$ at $\phi_L = \pi/2$
$\hat{S}$, $\hat{T}$	$\|1\rangle \to i\|1\rangle$, $\|1\rangle \to e^{i\pi/4}\|1\rangle$	quarter, eighth turn about z	virtual $\hat{Z}$
$\hat{R}_\phi(\theta)$	mixes $\|0\rangle, \|1\rangle$ with $\cos\frac{\theta}{2}$, $\sin\frac{\theta}{2}$	turn by θ about $(\cos\phi, \sin\phi, 0)$	area θ, phase $\phi_L = \phi$
$\hat{R}_z(\theta)$	$\|1\rangle \to e^{i\theta}\|1\rangle$ up to a phase	turn by θ about z	detuning for $t = \theta/\Delta$, or virtual $\hat{Z}$

The Hadamard gate is $\hat{H} = \hat{R}_y(\pi/2)\,\hat{Z}$, a virtual $\hat{Z}$ gate followed by one resonant pulse.

Gate Trajectories on the Bloch Sphere



We switch on $\hat{\mathcal{H}} = \frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ for $0 \le t \le 1$, and the evolution $\hat{U}(t) = \hat{R}_{\mathbf{n}}(\theta t)$ ends at the gate. The state starts at $\vartheta = 2\pi/3$, $\varphi = \pi/3$.

Each path is an arc around the axis of the gate and sweeps the rotation angle.

A One-Qubit Circuit

A **circuit** prepares a state, applies gates and measures. Each gate multiplies the state from the left, and the first gate stands rightmost in the matrix product.



matrix $\hat{S}\hat{H}$, $\hat{S}\hat{H}|0\rangle = \hat{S}|+\rangle = |+i\rangle$



matrix $\hat{H}\hat{S}$, $\hat{H}\hat{S}|0\rangle = \hat{H}|0\rangle = |+\rangle$

Interference, $\hat{H}\hat{Z}\hat{H} = \hat{X}$

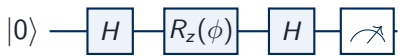
$\hat{H}$ exchanges the x and z axes, so a half turn about z between two Hadamard gates is a half turn about x,

$$|0\rangle \xrightarrow{\hat{H}} |+\rangle \xrightarrow{\hat{Z}} \underbrace{|-\rangle}_{\text{only the relative phase changed}} \xrightarrow{\hat{H}} |1\rangle.$$

The second $\hat{H}$ turns the phase difference into a different bit and thereby distinguishes $|+\rangle$ from $|-\rangle$.

Ramsey Interference

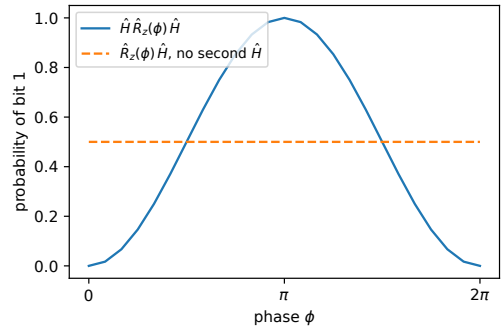
Replacing $\hat{Z}$ by $\hat{R}_z(\phi)$ gives the Ramsey circuit



with the amplitude of $|1\rangle$

$$\tfrac{1}{2}\left(e^{-i\phi/2} - e^{i\phi/2}\right) = -i \sin\tfrac{\phi}{2},$$

so $p(1) = \sin^2\frac{\phi}{2}$. Without the second $\hat{H}$, $p(1) = \frac{1}{2}$ for every ϕ, and the phase stays hidden.



With the closing $\hat{H}$ the probability oscillates between 0 and 1, and without it the record is a fair coin.

4

Other Bases and Shots



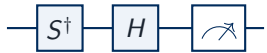
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Measurement Along x or y

A device reads out only the computational basis. To measure along another axis we first rotate that axis onto z.

$\hat{X}$ basis 

$\hat{H}|+\rangle = |0\rangle,\ \hat{H}|-\rangle = |1\rangle$

$\hat{Y}$ basis 

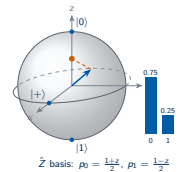
$\hat{S}^\dagger|\pm i\rangle = |\pm\rangle$, then $\hat{H}$

Bit 0 always stands for the eigenvalue $+1$ and bit 1 for -1, so in every basis

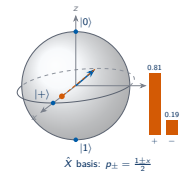
$$p_\alpha(0) = \frac{1 + \langle\hat{\sigma}_\alpha\rangle}{2}, \qquad p_\alpha(1) = \frac{1 - \langle\hat{\sigma}_\alpha\rangle}{2}, \qquad \alpha = x, y, z.$$

Running example

$\langle\hat{X}\rangle = \langle\hat{Y}\rangle = 0.612$ gives
$p_x(0) = p_y(0) = 0.806$, while
$p_z(0) = 0.750$.

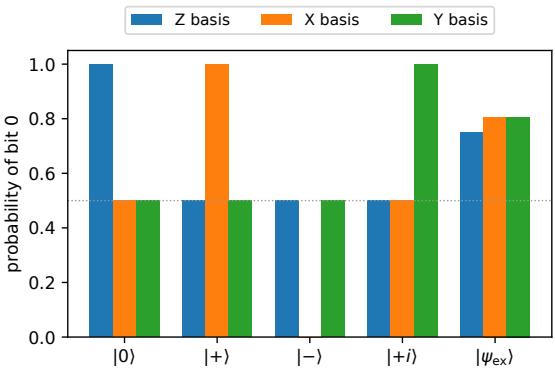


$\hat{Z}$ basis: $p_0 = \frac{1+z}{2}$, $p_1 = \frac{1-z}{2}$



$\hat{X}$ basis: $p_\pm = \frac{1\pm x}{2}$

Records in Different Bases



The bars show the probability of bit 0. A bar at 1 or 0 is a certain outcome, and a bar at 0.5 is a fair coin.

- No state gives a certain outcome in two bases at once (**complementarity**).
- $|+\rangle$ and $|-\rangle$ give identical $\hat{Z}$ records and opposite $\hat{X}$ records.
- ψ_{ex} leans to bit 0 in all three bases, since its Bloch vector lies in the octant $x, y, z > 0$.

Estimating an Expectation Value from Shots

M repetitions return $z_j = \pm 1$, and the sample mean estimates $\langle \hat{Z} \rangle$,

$$\bar{z} = \frac{1}{M} \sum_{j=1}^{M} z_j, \qquad \mathrm{Var}(z_j) = \underbrace{\langle \hat{Z}^2 \rangle}_{=1 \text{ since } z_j^2 = 1} - \langle \hat{Z} \rangle^2 .$$

For independent shots the standard error of the mean is

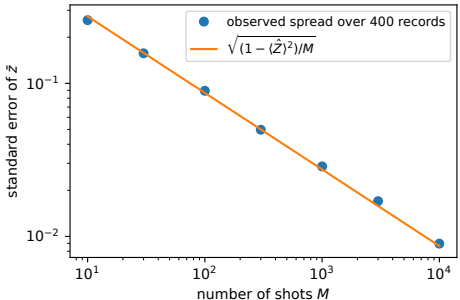
$$\mathrm{std}(\bar{z}) = \sqrt{\frac{1 - \langle \hat{Z} \rangle^2}{M}} .$$

Since $\langle \hat{Z} \rangle$ is unknown in practice, we divide the sample standard deviation of the record by $\sqrt{M}$. Doubling the precision costs four times as many shots.

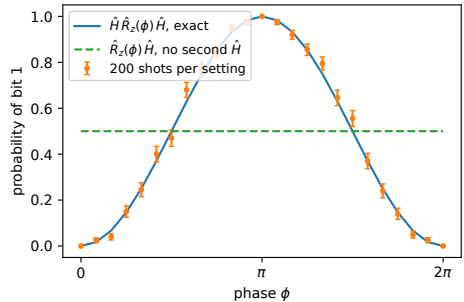
One record of 1000 shots of ψ_{ex}, exact $\langle \hat{Z} \rangle = 0.5$

The record gives $\bar{z} = 0.484 \pm 0.028$, and the formula gives $\sqrt{0.75/1000} = 0.027$.

Shot Noise and a Ramsey Experiment



Observed spread of the estimate over 400 repeated records against M, with slope $-1/2$ on log–log axes.



Ramsey circuit with 200 shots per phase. The error bars $\sqrt{p(1-p)/M}$ are largest near $p = \frac{1}{2}$.

Every expectation value measured on a device carries this error.

5

The Two-Spin Register in a Computer and Its Bit Strings



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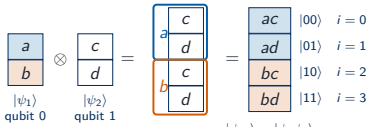
The Tensor Product

Qubit 0 in $|\psi_1\rangle$ and qubit 1 in $|\psi_2\rangle$ form the **product state** $|\psi_1\rangle \otimes |\psi_2\rangle$, with

$$|\psi_1\rangle = \begin{pmatrix} a \\ b \end{pmatrix} = a|0\rangle + b|1\rangle, \qquad |\psi_2\rangle = \begin{pmatrix} c \\ d \end{pmatrix} = c|0\rangle + d|1\rangle.$$

The product $\otimes$ is linear in each factor, so

$$|\psi_{12}\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \equiv |\psi_1\psi_2\rangle = ac\,\underbrace{|0\rangle \otimes |0\rangle}_{|00\rangle} + ad\,\underbrace{|0\rangle \otimes |1\rangle}_{|01\rangle} + bc\,\underbrace{|1\rangle \otimes |0\rangle}_{|10\rangle} + bd\,\underbrace{|1\rangle \otimes |1\rangle}_{|11\rangle}.$$



each entry of the first vector multiplies the whole second vector.
The first factor is qubit 0, the most significant bit

In code the product is `jnp.kron(psi1, psi2)`, and the norm factorises. The first factor, qubit 0, gives the left bit of the label. For $a = 1, b = 0$ and $c = d = \frac{1}{\sqrt{2}}$ we obtain $|0\rangle|+\rangle = \frac{1}{\sqrt{2}}(1, 1, 0, 0)^\top$.

Basis and Order of the 2^N Amplitudes

The four products of basis states span $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$, and a general state is a normalised superposition $|\psi\rangle = \sum_{b_0,b_1} \psi_{b_0 b_1} |b_0 b_1\rangle$ with $\sum |\psi_{b_0 b_1}|^2 = 1$. Most such states are not products. The label $b_0 b_1$ is a *bit string*, and the states $|b_0 b_1\rangle$ form the **computational basis**, the common eigenvectors of $\hat{Z}$ on each qubit with $+1$ for $b_q = 0$.

Stored order, the index of a bit string

A computer stores N qubits as 2^N complex amplitudes in this basis, in the order of the Kronecker product,

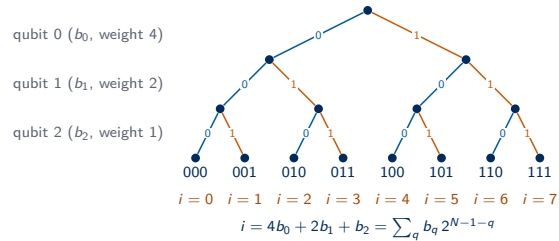
$$i = \sum_{q=0}^{N-1} b_q \, 2^{N-1-q} \qquad \text{(qubit 0 is the most significant bit)},$$

so qubit 0 changes slowest, as in counting in binary.

Ordering Tables and Memory

$N = 2$

bit string	i	$2b_0 + b_1$
00	0	$0 + 0$
01	1	$0 + 1$
10	2	$2 + 0$
11	3	$2 + 1$



- Rows and columns of operators use the same order.
- Some libraries, Qiskit among them, take qubit 0 as the least significant bit. Their bit strings read right to left and must be reordered before a comparison.
- At 16 bytes per complex amplitude the state takes $16 \cdot 2^N$ bytes, 16 GiB at $N = 30$, 16 TiB at $N = 40$ and 16 PiB at $N = 50$.

Kronecker Products of Operators

Operators on the register are Kronecker products as well. For 2×2 matrices the product is a 4×4 block matrix,

$$\hat{A} \otimes \hat{B} = \begin{pmatrix} A_{00}\hat{B} & A_{01}\hat{B} \\ A_{10}\hat{B} & A_{11}\hat{B} \end{pmatrix}, \qquad (\hat{A} \otimes \hat{B})_{2a+b,\, 2a'+b'} = A_{aa'} B_{bb'},$$

which acts factor by factor, $(\hat{A} \otimes \hat{B})(|u\rangle \otimes |v\rangle) = \hat{A}|u\rangle \otimes \hat{B}|v\rangle$. A gate $\hat{U}$ on qubit 0 alone is $\hat{U} \otimes \hat{I}$, and on qubit 1 alone $\hat{I} \otimes \hat{U}$, for example

$$\hat{X} \otimes \hat{I} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{matrix} 00 \leftrightarrow 10 \\ 01 \leftrightarrow 11 \end{matrix} \qquad \hat{I} \otimes \hat{X} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{matrix} 00 \leftrightarrow 01 \\ 10 \leftrightarrow 11 \end{matrix}$$

$\hat{X} \otimes \hat{I}$ flips the left bit and $\hat{I} \otimes \hat{X}$ the right bit. Diagonal operators read bits, $\hat{Z} \otimes \hat{I} = \mathrm{diag}(1, 1, -1, -1)$ and $\hat{I} \otimes \hat{Z} = \mathrm{diag}(1, -1, 1, -1)$, with $+1$ where $b_q = 0$. In code, `np.kron(X, I2)`.

Bit Strings of Two Qubits

For the register the Born rule gives the bit string $b_0 b_1$ with probability $p(b_0 b_1) = |\psi_{b_0 b_1}|^2$. A record of M shots is a list of M bit strings, summarised by three quantities.

- The **marginal** of qubit 0, $p_0(b_0) = \sum_{b_1} p(b_0 b_1)$, is what an observer of qubit 0 alone sees.
- The **single-qubit expectation** is $\langle \hat{Z}_q \rangle = p_q(0) - p_q(1)$.
- The **correlator** $\langle \hat{Z}_0 \hat{Z}_1 \rangle = \sum_{b_0 b_1} (-1)^{b_0 + b_1} p(b_0 b_1)$ equals $+1$ when the bits always agree, -1 when they always differ, and 0 when agreement and disagreement are equally likely.

Correlation and Independence

The bits are independent when the joint probabilities factorise, $p(b_0 b_1) = p_0(b_0)\, p_1(b_1)$. Independent bits have $\langle \hat{Z}_0 \hat{Z}_1 \rangle = \langle \hat{Z}_0 \rangle \langle \hat{Z}_1 \rangle$, which vanishes only when one of the two means is zero.

Product state $\hat{R}_y(2\pi/3)|0\rangle \otimes |0\rangle$ with qubit 0 tilted

Exact values are $\langle \hat{Z}_0 \rangle = -0.5$, $\langle \hat{Z}_1 \rangle = 1$ and $\langle \hat{Z}_0 \hat{Z}_1 \rangle = -0.5$, and 2000 shots give $\langle \hat{Z}_0 \hat{Z}_1 \rangle = -0.489 \pm 0.020$.

Bit Strings in Different Bases

A **measurement setting** assigns a basis to each qubit and is written as a pair of letters. In the setting $\hat{Z}\hat{X}$ qubit 0 is measured along z and qubit 1 along x, implemented as $\hat{H}$ on qubit 1 and then $\hat{Z}$ on both.

The state $|0\rangle \otimes |+\rangle$ has qubit 0 up along z and qubit 1 along $+x$. Counts in 1000 shots are

setting	00	01	10	11	
$\hat{Z}\hat{Z}$	508	492	0	0	qubit 0 fixed, qubit 1 random
$\hat{Z}\hat{X}$	1000	0	0	0	both fixed
$\hat{X}\hat{X}$	501	0	499	0	qubit 0 random, qubit 1 fixed

Only the setting that measures each qubit along its own axis gives a deterministic record. Since one state gives three different records, the setting is part of the experiment.

6

Two-Qubit Gates as Rotations



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Two-Qubit Gates from Interactions

Single-qubit gates map product states to product states. Correlations that no product state carries require an *interaction* of the spins. An Ising-type coupling, with energy proportional to the product of the same spin component $\alpha \in \{x, y, z\}$ of both qubits, gives

$$\hat{\mathcal{H}}_{\alpha\alpha} = \frac{J}{2}\,\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha \quad \Longrightarrow \quad \hat{R}_{\alpha\alpha}(\theta) = e^{-i\hat{\mathcal{H}}_{\alpha\alpha}t} = e^{-i\theta\,\hat{\sigma}_\alpha\otimes\hat{\sigma}_\alpha/2}, \qquad \theta = Jt.$$

Since $(\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha)^2 = \hat{\sigma}_\alpha^2 \otimes \hat{\sigma}_\alpha^2 = \hat{I}$, the identity $e^{-ia\hat{P}} = \cos a\,\hat{I} - i \sin a\,\hat{P}$ applies unchanged,

$$\hat{R}_{\alpha\alpha}(\theta) = \cos\tfrac{\theta}{2}\,\hat{I} - i \sin\tfrac{\theta}{2}\,\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha.$$

The angle θ is that of a rotation in the two-qubit space.

CNOT, CZ and SWAP

With the projectors $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ on qubit 0,

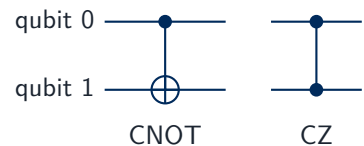
$$\widehat{\mathrm{CNOT}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{X},$$

$$\widehat{\mathrm{CZ}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{Z} = \mathrm{diag}(1, 1, 1, -1).$$

In the basis $00, 01, 10, 11$

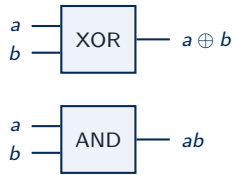
$$\widehat{\mathrm{CNOT}} = \begin{pmatrix} 1&0&0&0 \\ 0&1&0&0 \\ 0&0&0&1 \\ 0&0&1&0 \end{pmatrix}, \quad \widehat{\mathrm{SWAP}} = \begin{pmatrix} 1&0&0&0 \\ 0&0&1&0 \\ 0&1&0&0 \\ 0&0&0&1 \end{pmatrix}.$$

CNOT flips qubit 1 if and only if qubit 0 is 1, the reversible XOR $b_1 \to b_1 \oplus b_0$. CZ multiplies $|11\rangle$ by -1, and SWAP exchanges the qubits, $|01\rangle \leftrightarrow |10\rangle$.



The top wire is qubit 0, and time runs from left to right. The dot marks the **control**, and $\oplus$ the **target** that is flipped. CZ is symmetric in the two qubits and has a dot at both ends.

Reversible Classical Logic



two bits in, one out:
the inputs cannot be recovered



$00 \to 00$
$01 \to 01$
$10 \to 11$
$11 \to 10$

CNOT



$000 \to 000$ $100 \to 100$
$001 \to 001$ $101 \to 101$
$010 \to 010$ $110 \to 111$
$011 \to 011$ $111 \to 110$

Toffoli

reversible gates permute bit strings, so their matrices are permutation matrices, hence unitary;
Toffoli with $c = 0$ computes AND without losing the inputs

On bit strings CNOT computes XOR and Toffoli computes AND. Both are reversible, since the inputs can be recovered from the outputs, and as permutations of bit strings they are unitary. Every classical computation can be made reversible in this way, and a quantum computer can therefore run any classical program, with superposition and interference as additional resources.

Where Qubit Couplings Come From

Each platform has its own coupling, a sum of Pauli strings. iSWAP swaps $|01\rangle$, $|10\rangle$ with the phase i, and $\sqrt{\mathrm{SWAP}}$ applied twice is SWAP.

platform	mechanism	coupling	native gate
neutral atoms	Rydberg interaction $V \propto 1/R^6$	$V\|rr\rangle\langle rr\|$	controlled-Z by a blockade pulse sequence
trapped ions	Coulomb force, shared vibrations	$\hat{X}\hat{X}$	$\hat{R}_{xx}(\theta)$
superconducting circuits	capacitor or resonator	$\hat{X}\hat{X} + \hat{Y}\hat{Y}$, or $\hat{Z}\hat{X}$	iSWAP† or iSWAP, or CNOT
quantum-dot spins	exchange interaction	$J(t)\,\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1$	$\sqrt{\mathrm{SWAP}}$

Saffman *et al.*, RMP 82, 2313 (2010). Jaksch *et al.*, PRL 85, 2208 (2000). Sørensen, Mølmer, PRL 82, 1971 (1999). Rigetti, Devoret, PRB 81, 134507 (2010). Loss, DiVincenzo, PRA 57, 120 (1998)

Rydberg Interaction in Pauli Language

Projectors become Pauli operators, $|0\rangle\langle 0| = \frac{1}{2}(\hat{I} + \hat{Z})$ and $|1\rangle\langle 1| = \frac{1}{2}(\hat{I} - \hat{Z})$. With the qubit stored in $|g\rangle = |0\rangle$ and a Rydberg state $|r\rangle = |1\rangle$, the interaction of two atoms is

$$V|11\rangle\langle 11| = V\,\frac{\hat{I} - \hat{Z}_0}{2}\,\frac{\hat{I} - \hat{Z}_1}{2} = \underbrace{\tfrac{V}{4}\hat{Z}_0\hat{Z}_1}_{\text{Ising coupling}} \quad \underbrace{-\tfrac{V}{4}(\hat{Z}_0 + \hat{Z}_1)}_{\text{detuning of each atom}} \quad + \quad \underbrace{\tfrac{V}{4}\hat{I}}_{\text{constant}} .$$

The other couplings follow in the same way,

$$|01\rangle\langle 10| + |10\rangle\langle 01| = \tfrac{1}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1), \qquad \hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1 = \tfrac{1}{4}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1).$$

$\hat{R}_{xx}$ as a Rotation of the Pair $|00\rangle, |11\rangle$

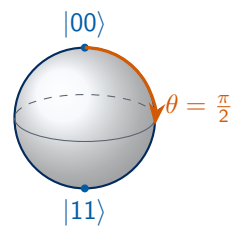
$\hat{X} \otimes \hat{X}$ flips both bits. With $c = \cos\frac{\theta}{2}$ and $s = \sin\frac{\theta}{2}$,

$$\hat{R}_{xx}(\theta) = c\hat{I} - is\,\hat{X} \otimes \hat{X} = \begin{pmatrix} c & 0 & 0 & -is \\ 0 & c & -is & 0 \\ 0 & -is & c & 0 \\ -is & 0 & 0 & c \end{pmatrix},$$

which acts as $\hat{R}_x(\theta) = \begin{pmatrix} c & -is \\ -is & c \end{pmatrix}$ on each pair $\{|00\rangle, |11\rangle\}$ and $\{|01\rangle, |10\rangle\}$. The pair $|00\rangle, |11\rangle$ spans a Bloch sphere with north pole $|00\rangle$ and south pole $|11\rangle$. At $\theta = \pi/2$ the rotation takes $|00\rangle$ to the equator,

$$\hat{R}_{xx}(\tfrac{\pi}{2})|00\rangle = \frac{|00\rangle - i|11\rangle}{\sqrt{2}},$$

which is no product state. Trapped ions get an $\hat{X}_0\hat{X}_1$ coupling from the Mølmer–Sørensen drive, and the drive duration sets θ.



Two Ions and Their Shared Motion

Two ions in one trap repel each other, and their joint oscillation is a **motional mode** of frequency ν, a harmonic oscillator shared by both ions.

- Its quanta are **phonons**, removed by $\hat{a}$ and created by $\hat{a}^\dagger$.
- A laser at the qubit frequency $\pm\nu$ flips a qubit and adds or removes one phonon, the blue and red **sidebands**.
- The **Lamb–Dicke parameter** $\eta \approx 0.1$, the motional spread over the reduced wavelength, sets the sideband strength.

Two tones detuned by $\pm(\nu - \delta)$ drive both ions. To first order in η they exert a force whose sign depends on the qubits,

$$\hat{\mathcal{H}}(t) = g\,\hat{S}_x\big(\hat{a}\,e^{-i\delta t} + \hat{a}^\dagger e^{i\delta t}\big), \qquad \hat{S}_x = \tfrac{1}{2}(\hat{X}_0 + \hat{X}_1), \quad g = \eta\Omega.$$

Sørensen, Mølmer, PRL 82, 1971 (1999). Mølmer, Sørensen, PRL 82, 1835 (1999)

A Closed Loop in Phase Space

For an eigenvalue $m \in \{1, 0, -1\}$ of $\hat{S}_x$ the motion is displaced by $m\,\alpha(t)$ with

$$\alpha(t) = \frac{g}{\delta}\left(1 - e^{i\delta t}\right),$$

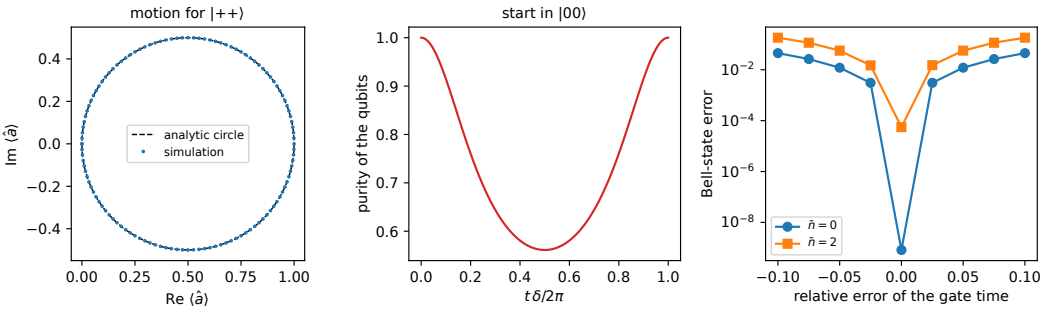
a circle that closes at $t = 2\pi K/\delta$ for every motional state. The loop leaves a phase $\propto m^2$, and with $\hat{S}_x^2 = \frac{1}{2}(\hat{I} + \hat{X}_0\hat{X}_1)$

$$\hat{U}\Big(\frac{2\pi K}{\delta}\Big) = e^{i\Phi\hat{S}_x^2} = e^{i\Phi/2}\,\hat{R}_{xx}(-\Phi), \qquad \Phi = \frac{2\pi K g^2}{\delta^2} = Jt, \quad J = \frac{g^2}{\delta}.$$

One loop with $\delta = 2g$ gives $\Phi = \pi/2$ and $|00\rangle \to (|00\rangle + i|11\rangle)/\sqrt{2}$. The opposite sign of δ reverses the rotation.

Sørensen, Mølmer, PRL 82, 1971 (1999). Mølmer, Sørensen, PRL 82, 1835 (1999)

A Gate Insensitive to Thermal Motion



The motion runs around the circle and returns (left). The qubits are entangled with the motion during the loop and pure again at the gate time (middle). The Bell-state fidelity is 0.9999 at mean phonon number 2. A gate time wrong by ten per cent gives the error 0.045 for cold and 0.182 for hot ions (right).

$\hat{R}_{zz}$ as a Conditional Rotation

$\hat{Z} \otimes \hat{Z}$ is diagonal. Writing $\hat{Z} = |0\rangle\langle 0| - |1\rangle\langle 1|$ for qubit 0 gives

$$\hat{Z} \otimes \hat{Z} = |0\rangle\langle 0| \otimes \hat{Z} - |1\rangle\langle 1| \otimes \hat{Z} \quad \Longrightarrow \quad \hat{R}_{zz}(\theta) = |0\rangle\langle 0| \otimes \hat{R}_z(\theta) + |1\rangle\langle 1| \otimes \hat{R}_z(-\theta),$$

$$\hat{R}_{zz}(\theta) = \mathrm{diag}\Big(\underbrace{e^{-i\theta/2},\, e^{i\theta/2}}_{\text{qubit } 0 = |0\rangle,\ \hat{R}_z(\theta)} \,,\quad \underbrace{e^{i\theta/2},\, e^{-i\theta/2}}_{\text{qubit } 0 = |1\rangle,\ \hat{R}_z(-\theta)} \Big).$$

- Qubit 1 turns about z by $+\theta$ or $-\theta$ depending on the state of qubit 0.
- The populations of the four bit strings never change, only their relative phases.
- On $|{+}{+}\rangle$ these conditional phases create a state that is not a product.
- Physically it is the Ising coupling $\frac{J}{2}\hat{Z}_0\hat{Z}_1$ on for $t = \theta/J$, from the Rydberg interaction or a tunable coupler.

CNOT and CZ as Conditional Half Turns

Since $\hat{X} = i\hat{R}_x(\pi)$, CNOT is a half turn about x of qubit 1, conditioned on qubit 0,

$$\widehat{\mathrm{CNOT}} = (\hat{S} \otimes \hat{I})\,[\,|0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_x(\pi)],$$

where $\hat{S}$ on qubit 0 supplies the phase i on the branch where qubit 0 is 1. As an evolution operator, let $\hat{P} = \underbrace{\frac{\hat{I}-\hat{Z}}{2}}_{|1\rangle\langle 1| \text{ on qubit 0}} \otimes \underbrace{\frac{\hat{I}-\hat{X}}{2}}_{|-\rangle\langle -| \text{ on qubit 1}}$, a projector with $\hat{P}^2 = \hat{P}$.

Summing the series with $\hat{P}^k = \hat{P}$ gives

$$e^{-i\pi\hat{P}} = \hat{I} + \underbrace{(e^{-i\pi} - 1)}_{-2}\hat{P} = \hat{I} - 2\hat{P},$$

which flips the sign of $|1\rangle|-\rangle$ alone, the action of CNOT. Hence

$$\widehat{\mathrm{CNOT}} = \exp[-i\pi(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{X})/4], \qquad \widehat{\mathrm{CZ}} = \exp[-i\pi(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{Z})/4].$$

CZ is generated by the projector on $|11\rangle$, and the sign in the exponent is irrelevant, since $e^{\pm i\pi} = -1$.

CZ and CNOT in the Laboratory

Expanding $(\hat{I} - \hat{Z}_0)(\hat{I} - \hat{Z}_1)$ in the exponent of CZ gives the Ising rotation and two z rotations,

$$\widehat{\mathrm{CZ}} = e^{-i\pi/4}\,[\hat{R}_z(-\tfrac{\pi}{2}) \otimes \hat{R}_z(-\tfrac{\pi}{2})]\,\hat{R}_{zz}(\tfrac{\pi}{2}), \qquad \widehat{\mathrm{CNOT}} = (\hat{I} \otimes \hat{H})\,\widehat{\mathrm{CZ}}\,(\hat{I} \otimes \hat{H}).$$

- CZ is the Ising coupling $\frac{J}{2}\hat{Z}_0\hat{Z}_1$ on for the calibrated time $t = \pi/(2J)$, with the two z rotations applied as virtual $\hat{Z}$ gates.
- Neutral atoms also reach CZ with a laser-pulse sequence that uses the Rydberg blockade.
- The Hadamard pair on the target turns CZ into CNOT, since $\hat{H}\hat{Z}\hat{H} = \hat{X}$. Superconducting circuits also obtain CNOT from the cross-resonance coupling $\hat{Z}\hat{X}$ on for a calibrated time, with single-qubit corrections.

SWAP and the Heisenberg Exchange

The exchange coupling $\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z}=4\,\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1$, with $\hat{\mathbf{S}}_q=\hat{\boldsymbol{\sigma}}_q/2$, is the interaction of neighbouring spins in a magnet.

eigenstate	exchange symmetry	eigenvalue of the coupling
$\lvert00\rangle,\ \lvert11\rangle,\ (\lvert01\rangle+\lvert10\rangle)/\sqrt{2}$	symmetric (triplet)	$+1$
$(\lvert01\rangle-\lvert10\rangle)/\sqrt{2}$	antisymmetric (singlet)	-3

SWAP is $+1$ on the symmetric states and -1 on the singlet, which gives both forms

$$\begin{aligned}\widehat{\mathrm{SWAP}} &= \tfrac{1}{2}\big(\hat{I}+\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z}\big)\\ &= e^{i\pi/4}\exp\big[-i\tfrac{\pi}{4}(\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z})\big].\end{aligned}$$

In the exponential form the symmetric states acquire the factor $e^{i\pi/4}e^{-i\pi/4}=1$, and the singlet acquires $e^{i\pi/4}e^{3i\pi/4}=-1$.

iSWAP and $\sqrt{\mathrm{SWAP}}$ from Exchange

Both gates act only on the pair $|01\rangle, |10\rangle$,

$$\mathrm{iSWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \sqrt{\mathrm{SWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1+i}{2} & \frac{1-i}{2} & 0 \\ 0 & \frac{1-i}{2} & \frac{1+i}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- The excitation exchange $\frac{J}{2}(\hat{X}\hat{X} + \hat{Y}\hat{Y})$ of superconducting circuits at $Jt = \pi/2$ gives iSWAP†, and iSWAP for the opposite sign of J.
- The Heisenberg exchange $J\,\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1$ of quantum-dot spins gives $\sqrt{\mathrm{SWAP}}$ for an exchange pulse with $\int J\,dt = \pi/2$, and SWAP for $\int J\,dt = \pi$.

Two-Qubit Angle from Interaction Time

A coupling $\frac{J(t)}{2}\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha$ commutes with itself at all times, so

$$\hat{U} = \hat{R}_{\alpha\alpha}(\theta), \qquad \theta = \int_0^T J(t')\,dt' \quad (= JT \text{ for a constant coupling}).$$

- $J(t)$ is set by a tunable coupler between superconducting qubits, by the strength and duration of the ion drive, or by the gate voltage of quantum-dot exchange.
- Any angle is reached by choosing the duration.
- Fixed gates such as CZ use a calibrated duration, measured once and reused.

Parametrised Gates and Their Physical Processes

A **parametrised gate** $\hat{U}(\theta)$ is a family labelled by a continuous angle. A **fixed gate** is one member at a calibrated angle.

gate	parameter	physical process	control setting
$\hat{R}_\phi(\theta)$	$\theta = \int \Omega\, dt$, axis ϕ_L	resonant laser or microwave pulse	duration or amplitude, phase
$\hat{R}_z(\theta)$, $\hat{S}$, $\hat{T}$	$\theta = \Delta t$ or phase shift	detuning, or virtual $\hat{Z}$	waiting time, or later phases
$\hat{R}_{zz}(\theta)$	$\theta = \int J\, dt$	Ising coupling, Rydberg or tunable coupler	interaction time
$\hat{R}_{xx}(\theta)$	$\theta = \int J\, dt$	Mølmer–Sørensen drive of ions	drive duration
$\sqrt{\mathrm{SWAP}}$, iSWAP	$\int J\, dt = \pi/2$	exchange of spins or circuits	exchange pulse $J(t)$
CZ, CNOT	$\hat{R}_{zz}(\pi/2)$ and fixed phases	Ising coupling, blockade, cross-resonance	calibrated duration

A Circuit as Timed Evolutions

Single-qubit drives and two-qubit couplings are both Hamiltonians switched on for a time. A circuit of K steps is

$$\hat{U} = e^{-it_K \hat{\mathcal{H}}_K} \cdots e^{-it_2 \hat{\mathcal{H}}_2} \, e^{-it_1 \hat{\mathcal{H}}_1}.$$

- Gates on different qubits run in parallel within a layer, and the circuit lasts the sum of its layer durations.
- Two-qubit gates take longer, as couplings between qubits are often weaker than the drives.

Each angle in a circuit diagram stands for the duration of a physical evolution.

7

Gates in an N-Qubit Register



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Pauli Strings

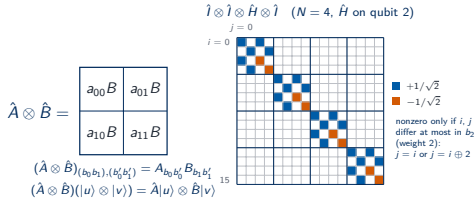
A **Pauli string** is a Kronecker product of one Pauli matrix or identity per qubit,

$$\hat{P}_0 \otimes \hat{P}_1 \otimes \cdots \otimes \hat{P}_{N-1}, \quad \hat{P}_q \in \{\hat{I}, \hat{X}, \hat{Y}, \hat{Z}\}, \qquad \text{for example } \hat{X}_0\hat{Z}_2 = \hat{X} \otimes \hat{I} \otimes \hat{Z} \otimes \hat{I} \ (N = 4).$$

- There are 4^N strings, each Hermitian with $\hat{P}^2 = \hat{I}$. Their eigenvalues are ± 1, and $e^{-ia\hat{P}} = \cos a\, \hat{I} - i \sin a\, \hat{P}$ applies.
- Two strings commute when the number of positions where both are non-identity and different is even, and anticommute otherwise.
- They form an operator basis with $\mathrm{Tr}(\hat{P}\hat{P}') = 2^N \delta_{PP'}$. Every Hamiltonian of N qubits is a real linear combination of strings, with the coefficient $\mathrm{Tr}(\hat{P}\hat{\mathcal{H}})/2^N$.
- Strings of $\hat{Z}$ and $\hat{I}$ alone are diagonal. $\hat{Z}_0\hat{Z}_2$ has the entry $(-1)^{b_0+b_2}$ at the bit string $b_0 b_1 b_2 b_3$, the parity of the selected bits.

A One-Qubit Gate on Qubit q of N

In Kronecker form the gate $\hat{U}$ stands at position q, with identities elsewhere. For $N = 4$ the Hadamard gate on qubit 2 is $\hat{H}_2 = \hat{I} \otimes \hat{I} \otimes \hat{H} \otimes \hat{I}$, a 16×16 matrix built by `reduce(np.kron, [I, I, H, I])`.



- It changes only bit 2, worth $2^{N-1-q} = 2$ in the index, and mixes each pair i, $i \oplus 2$, for example $\psi'_{0000} = (\psi_{0000} + \psi_{0010})/\sqrt{2}$ and $\psi'_{0010} = (\psi_{0000} - \psi_{0010})/\sqrt{2}$.
- Each row has two non-zero entries. The cost is 2^{N-1} pairs times four products, $2 \cdot 2^N$, against 4^N for a dense multiplication.

CNOT on a Non-Neighbouring Pair

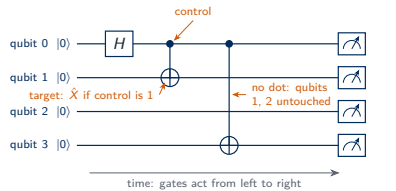
With control i, target j and the identity on every other qubit, the gate is a sum of two Kronecker products,

$$\widehat{\mathrm{CNOT}}_{i\to j} = |0\rangle\langle 0|_i \otimes \hat{I} + |1\rangle\langle 1|_i \otimes \hat{X}_j, \qquad \widehat{\mathrm{CNOT}}_{0\to 3} = |0\rangle\langle 0| \otimes \hat{I} \otimes \hat{I} \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{I} \otimes \hat{I} \otimes \hat{X}.$$

It permutes bit strings, $b_j \to b_j \oplus b_i$. For $N = 4$, $\widehat{\mathrm{CNOT}}_{0\to 3}$ acts as

before	after
1000 (8)	1001 (9)
1010 (10)	1011 (11)
1100 (12)	1101 (13)
1110 (14)	1111 (15)

and back, while strings with $b_0 = 0$ are unchanged.



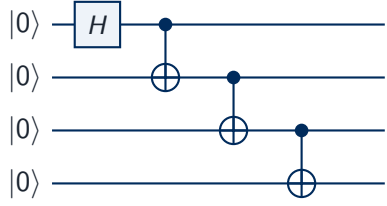
$\hat{U} = \mathrm{CNOT}_{0\to 3}\,\mathrm{CNOT}_{0\to 1}\,\hat{H}_0$ (matrices multiply from right to left)
top wire = qubit 0 = leftmost Kronecker factor = most significant bit

The GHZ State of N Qubits

The Greenberger–Horne–Zeilinger (GHZ) state of N qubits is

$$|\mathrm{GHZ}_N\rangle = \frac{|0\cdots0\rangle + |1\cdots1\rangle}{\sqrt{2}},$$

non-zero only at the indices 0 and $2^N - 1$. All bits agree in every shot of a $\hat{Z}$ measurement, while each bit alone is a fair coin.



A Hadamard gate on qubit 0 and the chain $\widehat{\mathrm{CNOT}}_{q-1\to q}$, $q = 1, \ldots, N-1$, prepare it. The fan-out $\widehat{\mathrm{CNOT}}_{0\to1}, \ldots, \widehat{\mathrm{CNOT}}_{0\to N-1}$ prepares the same state.

Sparse Operators

Most entries of $\hat{I} \otimes \cdots \otimes \hat{G} \otimes \cdots \otimes \hat{I}$ are zero. Code stores only the non-zero ones in compressed sparse rows (CSR), `scipy.sparse.kron(A, B, format="csr")`.

operator on N qubits	non-zero entries	dense entries
one-qubit gate $\hat{G}$ on qubit q	$2 \cdot 2^N$	4^N
Pauli string	2^N, one per row	4^N
$\widehat{\mathrm{CNOT}}_{i \to j}$ on any pair	2^N, a permutation	4^N
Hamiltonian with $\hat{L}$ Pauli strings	$\leq \hat{L} \cdot 2^N$	4^N

A matrix–vector product costs a time proportional to the number of non-zero entries, and at $N = 20$ it runs on a laptop. `scipy.sparse.linalg.expm_multiply` applies $e^{-it\hat{\mathcal{H}}}$ to $|\psi\rangle$ without forming the exponential. Sparse eigensolvers (Lanczos, `eigsh`) find ground states from products $\hat{\mathcal{H}}|\psi\rangle$ alone.

The State Tensor and `einsum`

A reshape of the 2^N vector into N axes of length 2 gives the **state tensor**, with one index per qubit, $\psi[b_0, \ldots, b_{N-1}] = \psi_i$ and $i = \sum_q b_q 2^{N-1-q}$. NumPy and JAX store arrays in C order, last axis fastest, so `psi.reshape((2,)*N)` moves no number.

Reshape and transpose

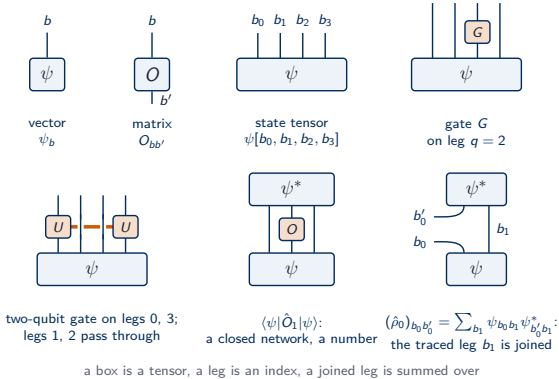
A reshape regroups, $(0,1,2,3) \to \psi[b_0, b_1] = \left(\begin{smallmatrix} 0 & 1 \\ 2 & 3 \end{smallmatrix}\right)$. A transpose exchanges, and flattening gives $(0,2,1,3)$, a different state with the qubits swapped.

`einsum` writes a sum over indices as letters. A letter in two inputs and absent from the output is summed, an output letter is kept, and output axes follow the order after `->`.

operation	`einsum`
$w_i = \sum_j A_{ij} v_j$	`"ij,j->i"`
$\langle a \vert b \rangle$	`"i,i->"` on `(a.conj(), b)`
$\mathrm{Tr} A$	`"ii->"`

Tensor Contractions

In the diagrams a box is a tensor, a line is an index, and a joined line is a summed index. A gate acts on the legs of its qubits, and the other legs pass through.



a box is a tensor, a leg is an index, a joined leg is summed over

- one-qubit gate on qubit 2 of 4, `"Cc,abcd->abCd"`
- two-qubit gate on legs 0 and 3, `"ADad,abcd->AbcD"`
- $\langle\psi|\hat{O}|\psi\rangle$, `"AB,ABab,ab->"`, a number with no open leg
- partial trace, `"ab,Ab->aA"`

The contractions give the same results as the Kronecker-product matrices at a cost proportional to 2^N per gate.

8

A Register of Interacting Qubits

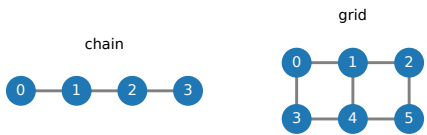


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A Register of Interacting Qubits

Each qubit of a register is driven on its own, and pairs joined by an edge of the **connectivity graph** interact,

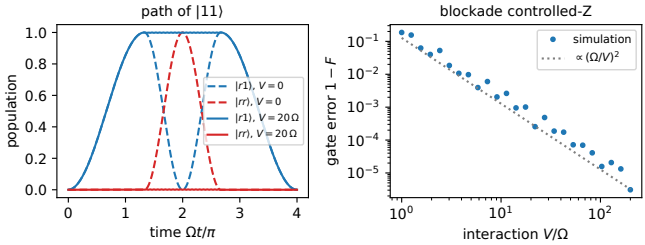
$$\hat{\mathcal{H}}_{\mathrm{dev}} = \sum_{q} \left(\frac{\Omega_q}{2} \hat{X}_q + \frac{\Delta_q}{2} \hat{Z}_q \right) + \sum_{(q,q') \in \mathrm{edges}} \frac{J_{qq'}}{2} \hat{Z}_q \hat{Z}_{q'}.$$



- Every term is a Pauli string, and pulses switch the terms on and off.
- A chain of 4 qubits has 11 strings and 80 nonzero entries out of 256.
- Distant qubits at distance d need $2(d-1)$ SWAP gates of three CNOTs each.

A Blockade Gate

The qubit is stored in two ground levels $|0\rangle$, $|1\rangle$, and $|r\rangle$ is a third, auxiliary level. A laser couples $|1\rangle \leftrightarrow |r\rangle$, and two atoms in $|r\rangle$ shift by V. The pulses π on the control, 2π on the target and π on the control give $\mathrm{diag}(1,-1,-1,-1) = (\hat{Z}\otimes\hat{Z})\,\widehat{\mathrm{CZ}}$, because V blocks the target in $|11\rangle$.



- At $V = 20\,\Omega$ the state $|rr\rangle$ holds at most 2.5×10^{-3}.
- A light shift of $|r1\rangle$ adds the phase $\pi\Omega/(2V)$, so the error is $\propto (\Omega/V)^2$, 1.3×10^{-3} at $V = 10\,\Omega$.
- An error of 10^{-3} needs $V \approx 11\,\Omega$.

D. Jaksch *et al.*, Phys. Rev. Lett. **85**, 2208 (2000), doi:10.1103/PhysRevLett.85.2208

9

Entanglement



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The Bell State and Its Predictions

The **Bell state** of two qubits is

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Each qubit alone

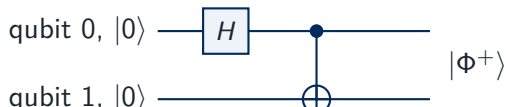
A measurement of one qubit in $\hat{Z}$, $\hat{X}$ or $\hat{Y}$ gives a fair coin, $\langle \hat{X}_q \rangle = \langle \hat{Y}_q \rangle = \langle \hat{Z}_q \rangle = 0$.

Both qubits together

The two bits agree in every shot in $\hat{Z}\hat{Z}$ and $\hat{X}\hat{X}$ and disagree in every shot in $\hat{Y}\hat{Y}$, $\langle \hat{Z}\hat{Z} \rangle = \langle \hat{X}\hat{X} \rangle = 1$, $\langle \hat{Y}\hat{Y} \rangle = -1$.

No product state $|u\rangle \otimes |v\rangle$ gives these records. Its probabilities factorise, $p(b_0 b_1) = p_0(b_0) p_1(b_1)$, and bits that always agree in $\hat{Z}$ would then require certain outcomes in p_0 and p_1, contradicting the fair coin of each qubit. The Bell state is therefore **entangled**, i.e. not a product of single-qubit states.

Preparing a Bell State



$$|00\rangle \xrightarrow{\hat{H}\otimes\hat{I}} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\widehat{\mathrm{CNOT}}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

CNOT flips qubit 1 only in the branch where qubit 0 is 1. The interaction gives the same kind of state, $\hat{R}_{xx}(\frac{\pi}{2})|00\rangle = (|00\rangle - i|11\rangle)/\sqrt{2}$, which $\hat{S}$ on qubit 0 maps to $|\Phi^+\rangle$.

The Product Test

We arrange the four amplitudes as a 2×2 matrix $\Psi_{b_0 b_1} = \psi_{b_0 b_1}$, with rows labelled by qubit 0. A product state has $\psi_{b_0 b_1} = a_{b_0} c_{b_1}$, so

$$\text{product state} \iff \Psi = a\, c^{\top} \iff \operatorname{rank} \Psi = 1.$$

$|0\rangle \otimes |+\rangle$

$$\Psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \operatorname{rank} \Psi = 1,$$

a product state.

$|\Phi^{+}\rangle$

$$\Psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \operatorname{rank} \Psi = 2,$$

an entangled state.

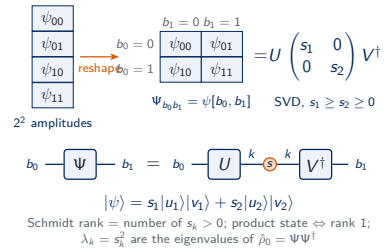
The Schmidt Decomposition

The singular value decomposition of Ψ is the **Schmidt decomposition** of $|\psi\rangle$,

$$\Psi = \sum_k s_k \, u_k v_k^\top \implies |\psi\rangle = \sum_{k=1}^{2} s_k \, |u_k\rangle \otimes |v_k\rangle,$$

with $s_1 \geq s_2 \geq 0$, $s_1^2 + s_2^2 = 1$ and orthonormal u_k, v_k. Measuring qubit 0 in the basis u_k gives the outcome k with probability s_k^2. The **Schmidt coefficients** s_k measure entanglement, and the number of non-zero s_k is the **Schmidt rank**, 1 for a product and 2 for an entangled state.

NumPy returns $\Psi = U \, \mathrm{diag}(s) \, V_h$, and v_k is the k-th row of V_h, without complex conjugation.



Schmidt Decompositions of the Two Examples

$|0\rangle \otimes |+\rangle$

$$\Psi = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \underbrace{1}_{s_1} \cdot \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{u_1 = |0\rangle} \underbrace{\tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \end{pmatrix}}_{v_1^\top = \langle + |}, \qquad s_2 = 0, \text{ a product.}$$

$|\Phi^+\rangle$

$$\Psi = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \tfrac{1}{\sqrt{2}}\,|0\rangle\langle 0| + \tfrac{1}{\sqrt{2}}\,|1\rangle\langle 1|, \qquad s_1 = s_2 = 0.7071, \text{ the largest spread.}$$

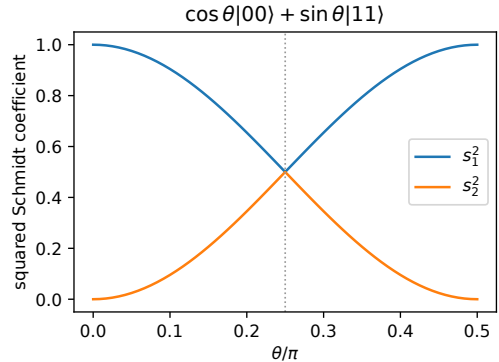
Gates on single qubits act as $\Psi \to U\Psi V^\top$ and keep the s_k, so $\hat{R}_{xx}(\frac{\pi}{2})|00\rangle$ and $|\Phi^+\rangle$ are equally entangled.

A Family of Product and Entangled States

$$|\psi_\theta\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle$$

has $\Psi = \mathrm{diag}(\cos\theta, \sin\theta)$ and the Schmidt coefficients $|\cos\theta|$ and $|\sin\theta|$.

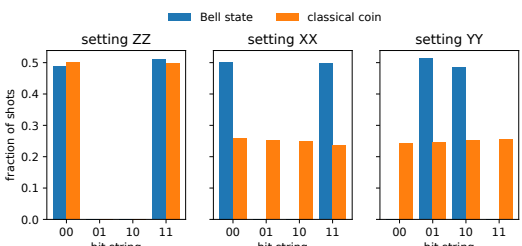
- At $\theta = 0$ the state is the product $|00\rangle$.
- At $\theta = \pi/4$ (dotted) it is the Bell state, with both squares equal to $\frac{1}{2}$.
- At $\theta = \pi/2$ it is the product $|11\rangle$, and beyond $\pi/4$ the two terms exchange roles.



The sorted squares $s_1^2 = \max(\cos^2\theta, \sin^2\theta)$ and $s_2^2 = \min(\cos^2\theta, \sin^2\theta)$.

Bell Correlations and a Classical Coin

A **classical correlated coin** prepares $|00\rangle$ or $|11\rangle$ with probability $\frac{1}{2}$ each and reproduces the $\hat{Z}\hat{Z}$ record of $|\Phi^+\rangle$.



Frequencies of the four bit strings in each setting, Bell state (blue) and coin (orange).

	$\langle \hat{Z}\hat{Z} \rangle$	$\langle \hat{X}\hat{X} \rangle$	$\langle \hat{Y}\hat{Y} \rangle$
Bell, exact	$+1$	$+1$	-1
Bell, sampled	$+1.000$	$+1.000$	-1.000
coin, sampled	$+1.000$	-0.006	-0.002

Since the coin holds $|00\rangle$ or $|11\rangle$ in every run, $\hat{X}$ and $\hat{Y}$ give fair, independent bits with $\langle \hat{X}\hat{X} \rangle = \langle \hat{Y}\hat{Y} \rangle = 0$. The samples use 4000 shots per setting.

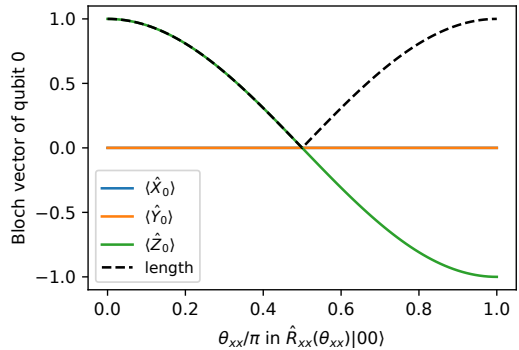
Entanglement Shrinks the Bloch Vector of Each Qubit

The Bloch vector of qubit 0 inside the register collects the expectation values of $\hat{X} \otimes \hat{I}$, $\hat{Y} \otimes \hat{I}$ and $\hat{Z} \otimes \hat{I}$,

$$\hat{R}_{xx}(\theta_{xx})|00\rangle = \cos\tfrac{\theta_{xx}}{2}|00\rangle - i \sin\tfrac{\theta_{xx}}{2}|11\rangle$$

$$\Longrightarrow \quad \mathbf{r}_0 = (0, 0, \cos\theta_{xx}).$$

Its length is one only for the products $|00\rangle$ and $|11\rangle$, and 0.000 at the Bell point $\theta_{xx} = \pi/2$, reached at $Jt = \pi/2$ for $\hat{\mathcal{H}} = \frac{J}{2}\hat{X} \otimes \hat{X}$. A vector inside the sphere describes no pure state of one qubit.



Components of $\mathbf{r}_0$ (solid) and its length $|\cos\theta_{xx}|$ (dashed) against the rotation angle of $\hat{R}_{xx}$. The x and y components stay zero.

10

The Register's Hamiltonian



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Hamiltonian of the Register

The register has Heisenberg exchange plus a magnetic field along z on both spins,

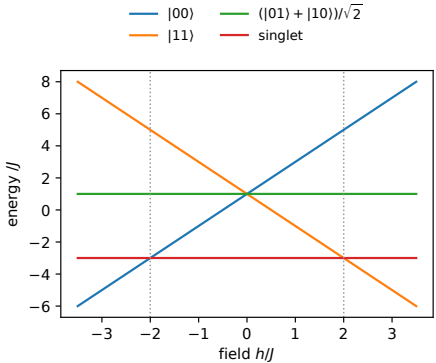
$$\hat{\mathcal{H}} = J\,(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1) + h\,(\hat{Z}_0 + \hat{Z}_1), \qquad \hat{X}_0\hat{X}_1 \equiv \hat{X}\otimes\hat{X},\quad \hat{Z}_0 \equiv \hat{Z}\otimes\hat{I}.$$

Each term is a Pauli string with a real coefficient. *Exact diagonalisation* of the 4×4 matrix gives all energies and stationary states, which match the triplet and singlet values of the exchange.

eigenstate	exchange	$\hat{Z}_0+\hat{Z}_1$	energy
$\lvert 00\rangle$	J	$+2$	$J+2h$
$\lvert 11\rangle$	J	-2	$J-2h$
$(\lvert 01\rangle+\lvert 10\rangle)/\sqrt{2}$	J	0	J
singlet $(\lvert 01\rangle-\lvert 10\rangle)/\sqrt{2}$	$-3J$	0	$-3J$

At $h=0$ the energies are $-3, 1, 1, 1$ in units of J. The ground state is the singlet, with $E=-3J$ and $s_1 = s_2 = 1/\sqrt{2}$.

Ground State and Magnetic Field



Energies against the field for $J > 0$, the *antiferromagnetic* sign, which favours anti-aligned spins.

- $(\hat{Z}_0 + \hat{Z}_1)$ gives zero on the singlet, whose level stays flat at $-3J$.
- $|11\rangle$ falls and crosses it at $h = 2J$, and $|00\rangle$ crosses at $h = -2J$ (dotted lines). A field $h > 0$ favours $|1\rangle$, opposite in sign to $-\frac{\omega_0}{2}\hat{Z}$ of the free qubit.
- For $|h| < 2J$ the ground state is the entangled singlet, and for $|h| > 2J$ a product state.

Hopping of an Excitation

Exchange without the $\hat{Z}\hat{Z}$ part,

$$\hat{\mathcal{H}}_{XX} = \tfrac{J}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1) = \tfrac{J}{4}(\hat{\sigma}_0^+\hat{\sigma}_1^- + \hat{\sigma}_0^-\hat{\sigma}_1^+)$$

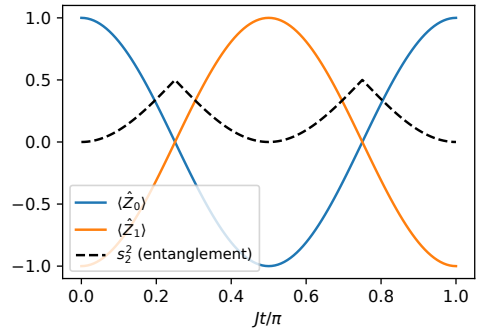
$$= J(|01\rangle\langle 10| + |10\rangle\langle 01|),$$

moves a flipped spin between the qubits. From $|01\rangle$

$$|\psi(t)\rangle = \cos(Jt)\,|01\rangle - i\sin(Jt)\,|10\rangle,$$

$$\langle \hat{Z}_0(t)\rangle = \cos 2Jt = -\langle \hat{Z}_1(t)\rangle.$$

The Schmidt coefficients $|\cos Jt|$ and $|\sin Jt|$ give maximal entanglement, $s_2^2 = 0.500$, at $Jt = \pi/4$ and $3\pi/4$.



The magnetisations oscillate in opposite phase. The dashed curve is the smaller squared Schmidt coefficient, maximal where they cross zero.

Couplings and Their Time Scales

The three couplings of the register carry different factors in front of their Pauli strings.

coupling	Hamiltonian	time scale
Ising $\hat{X}\hat{X}$	$\frac{J}{2}\,\hat{X}\otimes\hat{X}$, $\hat{R}_{xx}(\theta_{xx})$ with $\theta_{xx}=Jt$	Bell type from $\lvert 00\rangle$ at $Jt=\pi/2$
hopping	$\frac{J}{2}(\hat{X}\hat{X}+\hat{Y}\hat{Y})$	maximal entanglement from $\lvert 01\rangle$ at $Jt=\pi/4$
Heisenberg	$J(\hat{X}\hat{X}+\hat{Y}\hat{Y}+\hat{Z}\hat{Z})=4J\,\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1$	SWAP up to a phase at $Jt=\pi/4$

A rotation angle belongs to its Hamiltonian. Equal times t give different angles for different couplings.

Pauli Strings from Bit Strings

Bit 0 means the eigenvalue $+1$, so one shot gives the values

$$\hat{Z}_0 \to (-1)^{b_0}, \qquad \hat{Z}_1 \to (-1)^{b_1}, \qquad \hat{Z}_0\hat{Z}_1 \to \underbrace{(-1)^{b_0+b_1}}_{\text{parity of the bit string}} .$$

The mean over M shots estimates the expectation value. For a string with $\hat{X}$ or $\hat{Y}$ factors we first rotate those qubits with $\hat{H}$ or $\hat{H}\hat{S}^\dagger$ and then take the same parity.

One setting, several strings

The setting $\hat{Z}\hat{Z}$ gives $\hat{Z}_0$, $\hat{Z}_1$ and $\hat{Z}_0\hat{Z}_1$ from the same shots, all strings with $\hat{I}$ or the Pauli of the setting on each qubit. These strings commute.

setting	terms of $\hat{\mathcal{H}}$
$\hat{X}\hat{X}$	$\hat{X}_0\hat{X}_1$
$\hat{Y}\hat{Y}$	$\hat{Y}_0\hat{Y}_1$
$\hat{Z}\hat{Z}$	$\hat{Z}_0\hat{Z}_1,\ \hat{Z}_0,\ \hat{Z}_1$

Energy from Three Measurement Settings

With $J = 1$, $h = 0.5$ and 1000 shots per setting, the three parts of the energy come from independent settings, and their standard errors add in quadrature.

state	estimate from 3×1000 shots	exact
Bell $\lvert\Phi^+\rangle$	$+1.030 \pm 0.032$	$+1.000$
product $\lvert 0\rangle\lvert+\rangle$	$+0.590 \pm 0.065$	$+0.500$

- In the Bell state $\hat{X}_0\hat{X}_1$, $\hat{Y}_0\hat{Y}_1$ and $\hat{Z}_0\hat{Z}_1$ have the definite values $+1, -1, +1$, and exchange contributes J in every shot.
- In the Bell state the field part takes $\pm 2h$ with mean zero, since each qubit alone is random, and is the only source of error.
- $\lvert 0\rangle\lvert+\rangle$, with exact energy h, has random bits in more settings and a larger error.
- Both estimates agree with the exact energies within about 1.4 standard errors.

11

Density Matrices



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Density Matrix of a Pure State

The correlated coin and a qubit of the Bell state have no state vector. Every prediction for $|\psi\rangle$ is written with one matrix, $\hat{\rho} = |\psi\rangle\langle\psi|$, since the trace is cyclic, $\langle\hat{O}\rangle = \langle\psi|\hat{O}|\psi\rangle = \mathrm{Tr}(\hat{O}\,|\psi\rangle\langle\psi|) = \mathrm{Tr}(\hat{O}\hat{\rho})$. For one qubit, $|\psi\rangle = (a, b)^\top$,

$$\hat{\rho} = \begin{pmatrix} a \\ b \end{pmatrix} \begin{pmatrix} a^* & b^* \end{pmatrix} = \begin{pmatrix} |a|^2 & ab^* \\ a^*b & |b|^2 \end{pmatrix} \qquad \begin{matrix} \text{diagonal, populations } p(0), p(1) \\ \text{off-diagonal, coherences (relative phase)} \end{matrix}$$

The rank is 1, as every column is a multiple of $|\psi\rangle$, and $\hat{\rho}^2 = |\psi\rangle \underbrace{\langle\psi|\psi\rangle}_{=1}\langle\psi| = \hat{\rho}$, so

$\mathrm{Tr}\hat{\rho}^2 = 1$.

A product register has $\hat{\rho} = (|\psi_0\rangle \otimes |\psi_1\rangle)(\langle\psi_0| \otimes \langle\psi_1|) = \hat{\rho}_0 \otimes \hat{\rho}_1$, and for $|0\rangle|+\rangle$

$$\hat{\rho} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \tfrac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \text{rank } 1, \quad \hat{\rho}^2 = \hat{\rho}.$$

Mixtures

A source that emits $|\psi_i\rangle$ with probability p_i, without telling us which, is known only statistically. Its predictions average the Born predictions, $\langle\hat{O}\rangle = \sum_i p_i \langle\psi_i|\hat{O}|\psi_i\rangle = \mathrm{Tr}(\hat{O}\hat{\rho})$, with

$$\hat{\rho} = \sum_i p_i \, |\psi_i\rangle\langle\psi_i|, \qquad p_k = \langle k|\hat{\rho}|k\rangle.$$

- $\hat{\rho}$ is Hermitian with $\mathrm{Tr}\hat{\rho} = 1$ and positive, $\langle v|\hat{\rho}|v\rangle = \sum_i p_i |\langle v|\psi_i\rangle|^2 \geq 0$, so its eigenvalues λ_k are probabilities.
- The **purity** $\mathrm{Tr}\hat{\rho}^2 = \sum_k \lambda_k^2 \in [1/d, 1]$ equals 1 only for rank 1, a pure state. A mixture of distinct states has rank > 1 and purity < 1.

The Coin and the Bell State as Density Matrices

For the register, the classical coin and the Bell state have

$$\hat{\rho}_{\mathrm{Bell}} = \tfrac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \text{ (rank 1)}, \qquad \hat{\rho}_{\mathrm{coin}} = \tfrac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ (rank 2, } \mathrm{Tr}\hat{\rho}^2 = \tfrac{1}{2}\text{)}.$$

The two matrices share the diagonal and give identical $\hat{Z}\hat{Z}$ records. The coherences between $|00\rangle$ and $|11\rangle$ (orange) produce $\langle \hat{X}\hat{X} \rangle = 1$ and $\langle \hat{Y}\hat{Y} \rangle = -1$ for the Bell state and 0 for the coin.

Superposition and Mixture

We compare $|+\rangle$ with a source that emits $|0\rangle$ or $|1\rangle$ with probability $\frac{1}{2}$ each,

$$\hat{\rho}_{|+\rangle} = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \hat{\rho}_{\mathrm{mix}} = \tfrac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \tfrac{\hat{I}}{2}.$$

state	basis	exact $p(0), p(1)$	fraction of 0 in 2000 shots
$\|+\rangle$	$\hat{Z}$	0.5, 0.5	0.518
$\|+\rangle$	$\hat{X}$	1, 0	1.000
mixture	$\hat{Z}$	0.5, 0.5	0.495
mixture	$\hat{X}$	0.5, 0.5	0.523

The two states share the diagonal and hence the $\hat{Z}$ probabilities. The **coherences** (off-diagonal $\frac{1}{2}$) make the $\hat{X}$ outcome certain for $|+\rangle$, and the $\hat{X}$ basis tells the two states apart. One fixed basis sees only the diagonal in that basis and cannot determine an arbitrary density matrix.

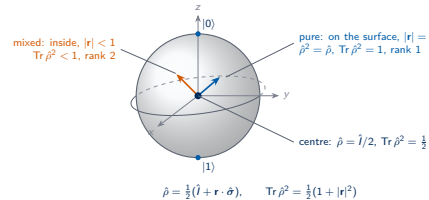
The Bloch Ball

Since $\hat{I}, \hat{X}, \hat{Y}, \hat{Z}$ are a basis of 2×2 matrices and $\mathrm{Tr}\hat{\rho} = 1$, three expectation values fix every density matrix of one qubit.

One qubit, $r_\alpha = \langle \hat{\sigma}_\alpha \rangle$

$$\hat{\rho} = \tfrac{1}{2}(\hat{I} + r_x \hat{X} + r_y \hat{Y} + r_z \hat{Z}),$$
$$|\mathbf{r}| \le 1, \qquad \mathrm{Tr}\hat{\rho}^2 = \tfrac{1}{2}(1 + |\mathbf{r}|^2).$$

The eigenvalues are $\frac{1}{2}(1 \pm |\mathbf{r}|)$, so pure states lie on the sphere and mixed states inside.



$\hat{I}/2$ sits at the centre with purity 0.5. A mixture averages Bloch vectors with its weights.

The Partial Trace

Let $|\psi\rangle = \sum_{a,b} \psi_{ab}|a\rangle|b\rangle$, with a for qubit 0 and b for qubit 1. A measurement on qubit 0 alone measures $\hat{O}_0 \otimes \hat{I}$, and with primed indices on the bra

$$\begin{aligned}\langle\psi|\hat{O}_0 \otimes \hat{I}|\psi\rangle &= \sum_{a,a',b,b'} \psi^*_{a'b'} \, (\hat{O}_0)_{a'a} \underbrace{\delta_{b'b}}_{\text{identity on qubit 1}} \psi_{ab} \\ &= \sum_{a,a'} (\hat{O}_0)_{a'a} \underbrace{\sum_b \psi_{ab}\psi^*_{a'b}}_{(\hat{\rho}_0)_{aa'}} = \mathrm{Tr}(\hat{O}_0\hat{\rho}_0).\end{aligned}$$

The Reduced Density Matrix

Reduced density matrix of qubit 0, the partial trace over qubit 1

$$(\hat{\rho}_0)_{aa'} = \sum_b \psi_{ab}\,\psi^*_{a'b}, \qquad \hat{\rho}_0 = \mathrm{Tr}_1|\psi\rangle\langle\psi|, \qquad (\hat{\rho}_1)_{bb'} = \sum_a \psi_{ab}\psi^*_{ab'}.$$

The delta sets the index of qubit 1 equal in ψ and ψ^*, and the sum over it is the sum over the unobserved bit. $\hat{\rho}_0$ contains everything that measurements on qubit 0 alone can reveal.

The Partial Trace Element by Element

For two qubits the four entries of $\hat{\rho}_0$ are

$$\hat{\rho}_0 = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{01}|^2 & \psi_{00}\psi_{10}^* + \psi_{01}\psi_{11}^* \\ \psi_{10}\psi_{00}^* + \psi_{11}\psi_{01}^* & |\psi_{10}|^2 + |\psi_{11}|^2 \end{pmatrix}.$$

- The diagonal holds the marginal probabilities of qubit 0, summed over both outcomes of qubit 1.
- The off-diagonal entries pair amplitudes that agree on qubit 1. If qubit 1 carries a record of qubit 0, these terms vanish.

The Partial Trace in Matrix Form

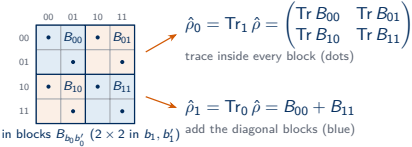
From the amplitudes, the reshape $\Psi_{ab} = \psi_{ab}$ with rows for qubit 0 turns the sum into a matrix product,

$$\hat{\rho}_0 = \Psi\Psi^\dagger, \qquad \hat{\rho}_1 = (\Psi^\dagger\Psi)^\top.$$

From the 4×4 density matrix, we cut $\hat{\rho}$ into four 2×2 blocks $\hat{\rho}^{(aa')}$ with rows $2a + b$, and

$$(\hat{\rho}_0)_{aa'} = \mathrm{Tr}\,\hat{\rho}^{(aa')}, \qquad \hat{\rho}_1 = \hat{\rho}^{(00)} + \hat{\rho}^{(11)}.$$

The block form holds for mixed registers as well. In code, `einsum("ab,Ab->aA", psi, psi.conj())`.



For the Bell state the block traces are $\frac{1}{2}, 0; 0, \frac{1}{2}$, so $\hat{\rho}_0 = \hat{I}/2$, and the sum of the diagonal blocks gives $\hat{\rho}_1 = \hat{I}/2$.

Three Reduced States

Product state $|0\rangle|+\rangle$

$\psi_{00} = \psi_{01} = \frac{1}{\sqrt{2}}$, $\psi_{10} = \psi_{11} = 0$, so $(\hat{\rho}_0)_{00} = \frac{1}{2} + \frac{1}{2} = 1$ and $\hat{\rho}_0 = |0\rangle\langle 0|$, a pure state.

Bell state

$\Psi = I/\sqrt{2}$ gives $\hat{\rho}_0 = \Psi\Psi^\dagger = \hat{I}/2$. The register is in a known pure state, while each qubit alone is a fair coin in every basis.

Family $\cos\theta|00\rangle + \sin\theta|11\rangle$

$\Psi = \mathrm{diag}(\cos\theta, \sin\theta)$ gives $\hat{\rho}_0 = \mathrm{diag}(\cos^2\theta, \sin^2\theta)$, which is $\mathrm{diag}(0.854,\ 0.146)$ at $\theta = \pi/8$.

The information of the Bell state sits in joint correlations. For Bell-diagonal states $\langle \hat{X}\hat{X}\rangle$, $\langle \hat{Y}\hat{Y}\rangle$, $\langle \hat{Z}\hat{Z}\rangle$ fix the register.

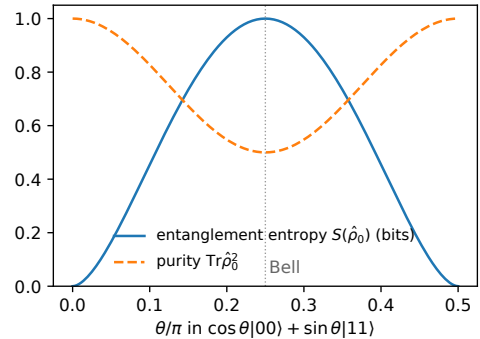
Purity and Entanglement Entropy

The **purity** $\mathrm{Tr}\hat{\rho}_0^2$ is 1 for the product, $\frac{1}{2}$ for the Bell state and $\cos^4\theta + \sin^4\theta$ for the family, 0.75 at $\theta = \pi/8$.

The **entanglement entropy** is the von Neumann entropy, in bits, of the eigenvalues λ_k of $\hat{\rho}_0$,

$$S(\hat{\rho}_0) = -\mathrm{Tr}\hat{\rho}_0 \log_2 \hat{\rho}_0 = -\sum_k \lambda_k \log_2 \lambda_k.$$

It is 0 for the product, 1 bit for the Bell state and $S_{\mathrm{bin}}(\cos^2\theta)$ for the family, 0.60 bits at $\theta = \pi/8$, with
$S_{\mathrm{bin}}(p) = -p\log_2 p - (1-p)\log_2(1-p)$.



The entropy rises to one bit at the Bell state while the purity falls to $\frac{1}{2}$.

Entanglement Criteria for a Pure Register

The SVD of the reshaped state, $\Psi = U\,\mathrm{diag}(s_k)\,V^\dagger$, gives

$$\hat{\rho}_0 = \Psi\Psi^\dagger = U\,\mathrm{diag}(s_k^2)\,U^\dagger \quad\Longrightarrow\quad \lambda_k = s_k^2.$$

$\Psi^\dagger\Psi$ has the same non-zero eigenvalues, so $S(\hat{\rho}_0) = S(\hat{\rho}_1)$ for a pure register, 0.36 bits on both sides for a random state.

pure register	Schmidt rank	$s_1,\ s_2$	$\mathrm{Tr}\hat{\rho}_0^2 = \sum_k s_k^4$	$S = -\sum_k s_k^2 \log_2 s_k^2$
product	1	$1,\ 0$	1	0
$\cos\theta\|00\rangle + \sin\theta\|11\rangle$	2 if $0 < \theta < \frac{\pi}{2}$	$\|\cos\theta\|,\ \|\sin\theta\|$	$\cos^4\theta + \sin^4\theta$	$S_{\mathrm{bin}}(\cos^2\theta)$
Bell	2	$\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}$	$\frac{1}{2}$	1

For a pure register, entangled $\Leftrightarrow$ Schmidt rank > 1 $\Leftrightarrow$ $\mathrm{Tr}\hat{\rho}_0^2 < 1$ $\Leftrightarrow$ $S(\hat{\rho}_0) > 0$.

Pure and Mixed, Global and Reduced

state	description	rank	purity	entropy (bits)
$\|0\rangle\|+\rangle$	pure product register	1	1	0
$\|\Phi^+\rangle$	pure entangled register	1	1	0
$\hat{\rho}_0$ of $\|\Phi^+\rangle$	reduced state of one qubit	2	1/2	1
$\hat{\rho}_{\text{coin}}$	classical mixture of $\|00\rangle, \|11\rangle$	2	1/2	1
$\hat{\rho}_0$ of $\hat{\rho}_{\text{coin}}$	reduced state of one qubit	2	1/2	1

Reduced entropy and mixed registers

The reduced entropy measures entanglement only when the global state is pure. Qubit 0 of the coin is the same $\hat{I}/2$ as qubit 0 of the Bell state, with one bit of entropy. For the coin, that bit is ignorance of which of $|00\rangle, |11\rangle$ was emitted. The coin contains no entanglement, and the two registers differ in $\langle \hat{X}\hat{X} \rangle$, $\langle \hat{Y}\hat{Y} \rangle$ and the global purity.

12

Measurement and Back-Action



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Projective Measurement

A projective measurement has one projector per outcome. The projectors are Hermitian, $\hat{P}_k = \hat{P}_k^\dagger$, orthogonal, $\hat{P}_k \hat{P}_l = \delta_{kl} \hat{P}_k$, so that outcomes exclude each other, and complete, $\sum_k \hat{P}_k = \hat{I}$, so that some outcome always occurs. The Born rule and the Lüders rule give

$$p_k = \mathrm{Tr}(\hat{P}_k \hat{\rho}), \qquad \hat{\rho} \longrightarrow \hat{\rho}_k = \frac{\hat{P}_k \, \hat{\rho} \, \hat{P}_k}{p_k}.$$

Measuring qubit 0 of the register

In the $\hat{Z}$ basis $\hat{P}_0 = |0\rangle\langle 0| \otimes \hat{I}$ and $\hat{P}_1 = |1\rangle\langle 1| \otimes \hat{I}$. In the $\hat{X}$ basis the projectors are $|\pm\rangle\langle\pm| \otimes \hat{I}$, with outcome bit 0 for $+$.

A pure state goes to $|\psi\rangle \to \hat{P}_k|\psi\rangle / \sqrt{p_k}$, the compatible branch renormalised.

G. Lüders, Ann. Phys. (Leipzig) **443**, 322 (1950), doi:10.1002/andp.19504430510

Back-Action on the Partner Qubit

For the Bell state, the outcome 0 of a $\hat{Z}$ measurement on qubit 0 gives

$$(|0\rangle\langle 0| \otimes \hat{I})\,\frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|00\rangle}{\sqrt{2}} \xrightarrow{\text{renormalise}} |0\rangle|0\rangle, \qquad p_0 = \tfrac{1}{2}.$$

In $\hat{X}$ the outcome $+$ leaves qubit 1 in $|+\rangle$ and the outcome $-$ in $|-\rangle$, since $|\Phi^+\rangle = (|{+}{+}\rangle + |{-}{-}\rangle)/\sqrt{2}$.

Qubit 0 is measured first and qubit 1 afterwards in the same basis, 200 shots each,

fraction of shots where qubit 1 repeats qubit 0	$\hat{Z}$	$\hat{X}$	$\hat{Y}$
Bell	1.00	1.00	0.00
classical coin	1.00	0.48	0.51

The partner of a Bell qubit is left in a definite eigenstate of the measured basis. It is the same one in $\hat{Z}$ and $\hat{X}$ and the opposite one in $\hat{Y}$, as $\langle \hat{Y}\hat{Y}\rangle = -1$. The coin agrees only in $\hat{Z}$.

No Signalling

Before the measurement qubit 1 of the Bell state is $\hat{I}/2$, a fair coin in every basis. After the measurement on qubit 0 it is in a definite pure state for an observer who knows the outcome.

At qubit 1 the outcome is unknown, and the post-measurement states are averaged,

$$\hat{\rho}_1^{\text{after}} = \sum_k p_k\, \hat{\rho}_{1,k} = \tfrac{1}{2}|0\rangle\langle 0| + \tfrac{1}{2}|1\rangle\langle 1| = \frac{\hat{I}}{2} \quad (\hat{Z}), \qquad \tfrac{1}{2}|+\rangle\langle +| + \tfrac{1}{2}|-\rangle\langle -| = \frac{\hat{I}}{2} \quad (\hat{X}).$$

- For any basis chosen at qubit 0, qubit 1 alone remains $\hat{I}/2$, as the simulation confirms for $\hat{Z}$, $\hat{X}$ and $\hat{Y}$.
- The correlations appear only when the two records are brought together.
- The basis chosen at qubit 0 cannot be read off at qubit 1 (**no signalling**).

Unread Measurement and Dephasing

A measurement whose result is not recorded maps

$$\hat{\rho} \;\longrightarrow\; \sum_k p_k \hat{\rho}_k = \sum_k \hat{P}_k \,\hat{\rho}\, \hat{P}_k .$$

It keeps the blocks of $\hat{\rho}$ inside each outcome sector and removes the coherences between sectors. For a measurement in a basis it keeps the diagonal and removes all coherences.

One qubit

An unread $\hat{Z}$ measurement on $|+\rangle$ gives $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \to \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, with purity $1 \to 0.5$.

Register

An unread $\hat{Z}$ measurement of qubit 0 turns $\hat{\rho}_{\mathrm{Bell}}$ into $\hat{\rho}_{\mathrm{coin}}$, with $\langle \hat{X}\hat{X} \rangle \; 1 \to 0$ and $\langle \hat{Z}\hat{Z} \rangle \; 1 \to 1$.

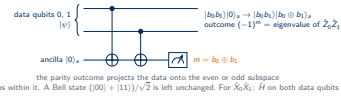
The loss of coherence in a fixed basis is called **dephasing**, and an environment that records the qubit has the same effect.

Measuring a Pauli String as a Parity

$\hat{Z}_0\hat{Z}_1$ has the eigenvalue $(-1)^{b_0 \oplus b_1}$, and measuring it reveals only the *parity* $b_0 \oplus b_1$, with the projectors $\hat{P}_{\text{even}} = |00\rangle\langle 00| + |11\rangle\langle 11|$ and $\hat{P}_{\text{odd}} = |01\rangle\langle 01| + |10\rangle\langle 10|$.

A readout of both bits, after rotating $\hat{X}$ or $\hat{Y}$ factors to $\hat{Z}$, and multiplying the ± 1 values gives the parity. It also reveals the individual bits and destroys superpositions inside each parity sector.

CNOT gates from qubits 0 and 1 onto an extra qubit (ancilla) in $|0\rangle$ write $b_0 \oplus b_1$ into it. Measuring the ancilla alone is the projective measurement $\{\hat{P}_{\text{even}}, \hat{P}_{\text{odd}}\}$. A Bell state gives even parity with certainty and is left unchanged.



The same circuit measures stabilisers in quantum error correction.

A Noisy Detector

A $\hat{Z}$ detector that reports the wrong bit with probability ϵ gives the outcome probabilities

$$p(0) = \mathrm{Tr}(\hat{\pi}_0 \hat{\rho}), \qquad \hat{\pi}_0 = (1-\epsilon)|0\rangle\langle 0| + \epsilon |1\rangle\langle 1|, \qquad \hat{\pi}_1 = \hat{I} - \hat{\pi}_0.$$

The operators $\hat{\pi}_0$ and $\hat{\pi}_1$ are positive and sum to the identity. Since $\hat{\pi}_0^2 \neq \hat{\pi}_0$ for $0 < \epsilon < 1$, they are not projectors.

With $\epsilon = 0.1$ the state $|0\rangle$ gives $p(0) = 0.9$, and $|+\rangle$ gives $p(0) = 0.5$. The same numbers form the confusion matrix of a readout error, which maps the true probabilities of the bits to the recorded ones.

General Measurements

A **positive operator-valued measure** (POVM) is a set $\{\hat{\pi}_i\}$ with $\hat{\pi}_i \geq 0$, so that probabilities are non-negative, and $\sum_i \hat{\pi}_i = \hat{I}$, so that they sum to one. The Born rule reads $p(i) = \mathrm{Tr}(\hat{\pi}_i \hat{\rho})$, and the $\hat{\pi}_i$ need be neither projectors nor orthogonal.

The POVM fixes only the statistics. The state after the outcome i follows from an **instrument**, Kraus operators $\hat{M}_{i,\alpha}$ with

$$\hat{\pi}_i = \sum_\alpha \hat{M}_{i,\alpha}^\dagger \hat{M}_{i,\alpha}, \qquad \hat{\rho} \to \hat{\rho}^{(i)} = \frac{1}{p(i)} \sum_\alpha \hat{M}_{i,\alpha}\, \hat{\rho}\, \hat{M}_{i,\alpha}^\dagger.$$

Different instruments can realise the same POVM and leave different states. One example is the **square-root instrument** $\hat{M}_i = \sqrt{\hat{\pi}_i}$, with $\hat{\rho}^{(i)} = \sqrt{\hat{\pi}_i}\hat{\rho}\sqrt{\hat{\pi}_i}/p(i)$, and a projective measurement has $\hat{\pi}_i = \hat{M}_i = \hat{P}_i$.

Informationally Complete Measurements

A measurement is **informationally complete** if its outcome probabilities fix $\hat{\rho}$. Its POVM elements must span all Hermitian $d \times d$ matrices, $d = 2^N$, so an informationally complete POVM has at least d^2 outcomes.

Pauli settings

Measuring every qubit in $\hat{X}$, $\hat{Y}$ or $\hat{Z}$ gives 3^N settings. Their parities give all 4^N Pauli-string expectation values, and

$$\hat{\rho} = \frac{1}{2^N} \sum_P \langle \hat{P} \rangle \, \hat{P}.$$

When the setting is chosen at random, it becomes part of the outcome label, and each element carries the probability of its setting.

State tomography is limited by the number of settings, which grows exponentially with N.

A Symmetric Informationally Complete POVM

A symmetric informationally complete POVM (SIC-POVM) of one qubit uses the four corners of a regular tetrahedron in the Bloch ball,

$$\mathbf{s}_1 = (0, 0, 1), \quad \mathbf{s}_2 = (\tfrac{2\sqrt{2}}{3}, 0, -\tfrac{1}{3}), \quad \mathbf{s}_{3,4} = (-\tfrac{\sqrt{2}}{3}, \pm\sqrt{\tfrac{2}{3}}, -\tfrac{1}{3}),$$

with the elements

$$\hat{\pi}_a = \tfrac{1}{4}(\hat{I} + \mathbf{s}_a \cdot \hat{\boldsymbol{\sigma}}), \qquad a = 1, \ldots, 4.$$

- $\sum_a \hat{\pi}_a = \hat{I}$, and all pairs of elements overlap equally, $\mathrm{Tr}\hat{\pi}_a\hat{\pi}_b = \frac{1}{12}$ for $a \neq b$.
- One setting returns four outcomes, and their probabilities determine the Bloch vector.
- For N qubits the products $\hat{\pi}_{a_0} \otimes \cdots \otimes \hat{\pi}_{a_{N-1}}$ give 4^N outcomes.

J. M. Renes, R. Blume-Kohout, A. J. Scott and C. M. Caves, J. Math. Phys. **45**, 2171 (2004), doi:10.1063/1.1737053

13

Qubit Errors, Noise Channels and Distances



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An Environment That Records the Qubit

Noise is a coupling to an unobserved environment E. In the smallest example one environment qubit starts in $|0\rangle_E$, and the coupling $\hat{\mathcal{H}} = \frac{\Omega}{2}\,|1\rangle\langle 1| \otimes \hat{Y}$ rotates it about y only when the system is $|1\rangle$,

$$\hat{U} = e^{-i\hat{\mathcal{H}}t} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_y(g), \qquad g = \Omega t, \quad 0 \le g \le \pi,$$
$$|0\rangle|0\rangle_E \to |0\rangle|0\rangle_E, \qquad |1\rangle|0\rangle_E \to |1\rangle\big(\cos\tfrac{g}{2}|0\rangle_E + \sin\tfrac{g}{2}|1\rangle_E\big).$$

Collecting the terms with $|0\rangle_E$ and with $|1\rangle_E$ leaves a 2×2 matrix on the system for each,

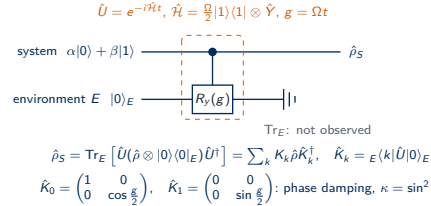
$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \cos\frac{g}{2} \end{pmatrix}, \qquad \hat{K}_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sin\frac{g}{2} \end{pmatrix}, \qquad \mathcal{E}(\hat{\rho}) = \begin{pmatrix} \rho_{00} & \cos\frac{g}{2}\,\rho_{01} \\ \cos\frac{g}{2}\,\rho_{10} & \rho_{11} \end{pmatrix}.$$

The populations are unchanged, and the coherence is multiplied by $\cos\frac{g}{2}$, since the environment has partly recorded whether the system is in $|1\rangle$. The channel is *phase damping* with $\kappa = \sin^2\frac{g}{2}$, and $g = \pi$ gives a full unread measurement. For $g > \pi$ the overlap $\cos\frac{g}{2}$ changes sign and adds a phase flip.

Kraus Operators

For any unitary coupling $\hat{U}$ we trace out E with the partial trace Tr_E,

$$\begin{aligned}\mathcal{E}(\hat{\rho}) &= \mathrm{Tr}_E\big[\hat{U}(\hat{\rho}\otimes|0\rangle\langle 0|_E)\hat{U}^\dagger\big] \\ &= \sum_k {}_E\langle k|\hat{U}|0\rangle_E\,\hat{\rho}\,{}_E\langle 0|\hat{U}^\dagger|k\rangle_E \\ &= \sum_k \hat{K}_k\hat{\rho}\hat{K}_k^\dagger.\end{aligned}$$



The **Kraus operator** $\hat{K}_k = {}_E\langle k|\hat{U}|0\rangle_E$ takes the matrix element of $\hat{U}$ in the environment alone, from the input $|0\rangle_E$ to the output $|k\rangle_E$, and leaves a system matrix. Inserting $\sum_k |k\rangle\langle k|_E = \hat{I}_E$ gives

$$\sum_k \hat{K}_k^\dagger\hat{K}_k = {}_E\langle 0|\hat{U}^\dagger\hat{U}|0\rangle_E = \hat{I} \quad\Longrightarrow\quad \mathrm{Tr}\,\mathcal{E}(\hat{\rho}) = \mathrm{Tr}\hat{\rho} = 1.$$

K. Kraus, *States, Effects, and Operations*, Lect. Notes Phys. **190** (Springer, 1983), doi:10.1007/3-540-12732-1

Pauli Errors and Depolarising Noise

With probability p a Pauli error $\hat{\sigma}$ hits the qubit, $\hat{K}_0 = \sqrt{1-p}\,\hat{I}$, $\hat{K}_1 = \sqrt{p}\,\hat{\sigma}$, and $\mathcal{E}(\hat{\rho}) = (1-p)\,\hat{\rho} + p\,\hat{\sigma}\hat{\rho}\hat{\sigma}$.

error	Kraus operators	Bloch vector $(r_x, r_y, r_z) \to$
bit flip	$\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{X}$	$\big(r_x,\ (1-2p)r_y,\ (1-2p)r_z\big)$
phase flip (dephasing)	$\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{Z}$	$\big((1-2p)r_x,\ (1-2p)r_y,\ r_z\big)$
bit-phase flip	$\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{Y}$	$\big((1-2p)r_x,\ r_y,\ (1-2p)r_z\big)$
depolarising	$\sqrt{1-p}\,\hat{I},\ \sqrt{p/3}\,\hat{X},\ \sqrt{p/3}\,\hat{Y},\ \sqrt{p/3}\,\hat{Z}$	$\big(1-\frac{4}{3}p\big)\,\mathbf{r}$

Anticommutation gives $\hat{\sigma}\hat{\sigma}_\alpha\hat{\sigma} = +\hat{\sigma}_\alpha$ for $\hat{\sigma}_\alpha = \hat{\sigma}$ and $-\hat{\sigma}_\alpha$ otherwise, so the component along $\hat{\sigma}$ survives and the other two shrink by $1-2p$. Depolarising noise is $\mathcal{E}(\hat{\rho}) = (1-\frac{4}{3}p)\hat{\rho} + \frac{4}{3}p\,\frac{\hat{I}}{2}$, a uniform shrink, and at $p = \frac{3}{4}$ every state becomes $\hat{I}/2$.

Amplitude Damping and Phase Damping

Amplitude damping, energy loss

$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad \hat{K}_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}$$

$\hat{K}_1 = \sqrt{\gamma}\,\hat{S}^+$ takes $|1\rangle \to |0\rangle$ with probability γ,

$$\mathcal{E}(\hat{\rho}) = \begin{pmatrix} \rho_{00} + \gamma\rho_{11} & \sqrt{1-\gamma}\,\rho_{01} \\ \sqrt{1-\gamma}\,\rho_{10} & (1-\gamma)\rho_{11} \end{pmatrix}.$$

$r_{x,y} \to \sqrt{1-\gamma}\, r_{x,y}$, $r_z \to \gamma + (1-\gamma) r_z$, towards the fixed state $|0\rangle$.

Phase damping, no energy loss

$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\kappa} \end{pmatrix}, \quad \hat{K}_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\kappa} \end{pmatrix}$$

The environment learns about $|1\rangle$ and takes no energy,

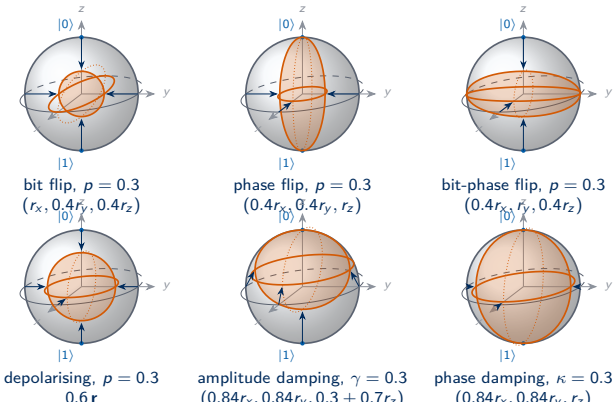
$$\mathcal{E}(\hat{\rho}) = \begin{pmatrix} \rho_{00} & \sqrt{1-\kappa}\,\rho_{01} \\ \sqrt{1-\kappa}\,\rho_{10} & \rho_{11} \end{pmatrix}.$$

$r_{x,y} \to \sqrt{1-\kappa}\, r_{x,y}$, r_z unchanged.

Phase damping equals the phase flip with $1 - 2p = \sqrt{1-\kappa}$.

Six Error Models on the Bloch Ball

Each panel shows the image (orange) of the Bloch sphere (grey) under one channel at strength $p = \gamma = \kappa = 0.3$, with arrows from pure states to their images.



Pauli errors squeeze the ball towards the axis of the error, and depolarising noise shrinks it uniformly by $1 - \frac{4}{3}p = 0.6$. Only amplitude damping moves the centre, towards $|0\rangle$. Phase damping keeps the poles and shrinks x and y by $\sqrt{1-\kappa} \approx 0.84$.

Quantum Channels

Every Kraus map $\mathcal{E}(\hat{\rho}) = \sum_k \hat{K}_k \hat{\rho} \hat{K}_k^\dagger$ with $\sum_k \hat{K}_k^\dagger \hat{K}_k = \hat{I}$ is **completely positive**, since it keeps $\hat{\rho} \geq 0$ even when it acts on one qubit of a larger register ($\mathcal{E} \otimes \mathrm{id}$). It is also **trace preserving**, $\mathrm{Tr}\, \mathcal{E}(\hat{\rho}) = 1$. Such a completely positive trace-preserving (CPTP) map, a quantum channel, is the most general physical evolution, and a unitary is the case of one Kraus operator.

Pauli channels

$\mathcal{E}(\hat{\rho}) = p_0 \hat{\rho} + p_x \hat{X} \hat{\rho} \hat{X} + p_y \hat{Y} \hat{\rho} \hat{Y} + p_z \hat{Z} \hat{\rho} \hat{Z}$ with $\sum p = 1$. Bit, phase and bit-phase flip and depolarising noise are special cases.

Unital and non-unital channels

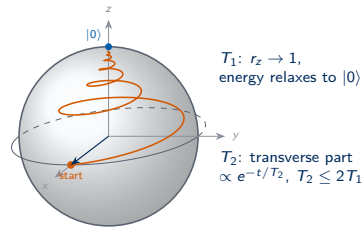
A unital channel has $\mathcal{E}(\hat{I}) = \hat{I}$ and keeps the centre of the Bloch ball fixed, as Pauli channels and phase damping do. Amplitude damping is non-unital and drives every state towards $|0\rangle$.

The Lindblad Master Equation

For a qubit weakly coupled to a memoryless environment the density matrix obeys

$$\frac{d\hat{\rho}}{dt} = \underbrace{-i[\hat{\mathcal{H}}, \hat{\rho}]}_{\text{unitary part}} + \sum_j \underbrace{\left(\hat{L}_j \hat{\rho} \hat{L}_j^\dagger - \tfrac{1}{2}\{\hat{L}_j^\dagger \hat{L}_j, \hat{\rho}\}\right)}_{\text{jump operator } \hat{L}_j}.$$

The solution over a time t is a channel. Energy decay with $\hat{L} = \sqrt{\Gamma}\,|g\rangle\langle e| = \sqrt{\Gamma}\,\hat{S}^+$ and $T_1 = 1/\Gamma$ gives amplitude damping, $\gamma = 1 - e^{-t/T_1}$. Pure dephasing with $\hat{L} = \sqrt{\gamma_\varphi/2}\,\hat{Z}$ gives phase damping, $\sqrt{1-\kappa} = e^{-\gamma_\varphi t}$. With both, $\rho_{11} \propto e^{-t/T_1}$ and $\rho_{01} \propto e^{-t/T_2}$, $\frac{1}{T_2} = \frac{1}{2T_1} + \gamma_\varphi$.

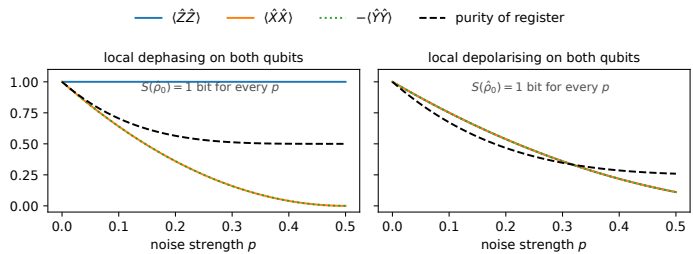


With detuning the Bloch vector precesses while it relaxes.

G. Lindblad, Commun. Math. Phys. **48**, 119 (1976), doi:10.1007/BF01608499

The Register under Local Noise

The same channel acts on each qubit of the Bell state, $\hat{\rho} \to \sum_{k,l} (\hat{K}_k \otimes \hat{K}_l)\, \hat{\rho}\, (\hat{K}_k \otimes \hat{K}_l)^\dagger$.



Dephasing keeps $\langle \hat{Z}\hat{Z} \rangle = 1$ and shrinks $\langle \hat{X}\hat{X} \rangle$ to $(1-2p)^2$, 0.16 at $p = 0.3$, and at $p = \frac{1}{2}$ the register becomes the classical coin. Depolarising noise shrinks all correlators by $(1 - \frac{4}{3}p)^2$, 0.36 at $p = 0.3$.

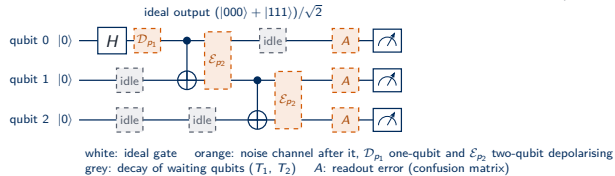
Entropy of One Qubit under Local Noise

Under local dephasing and depolarising noise the reduced entropy $S(\hat{\rho}_0)$ (grey dots) stays at 1 bit for every p, since $\hat{\rho}_0 = \hat{I}/2$ throughout. The register purity at $p = 0.3$ under dephasing is 0.51.

The loss of entanglement therefore escapes the reduced entropy. It shows in the correlators $\langle \hat{X}\hat{X} \rangle$ and $\langle \hat{Y}\hat{Y} \rangle$ and in the global purity, and each of these requires joint measurements.

The Gate-Noise Model of a Circuit

Each ideal gate is followed by a channel on the qubits it touched, $\hat{\rho} \to \mathcal{E}(\hat{U}\hat{\rho}\hat{U}^\dagger)$.

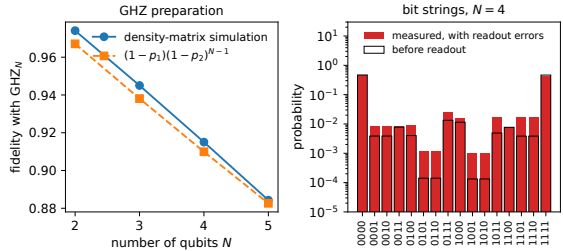


- Depolarising noise acts with p_1 after one-qubit gates and with p_2 after each CNOT, $\hat{\rho} \to (1-p_2)\hat{\rho} + \frac{p_2}{15}\sum_{\hat{P}\neq\hat{I}\hat{I}} \hat{P}\hat{\rho}\hat{P}$, typically with $p_2 \approx 10\,p_1$.
- Qubits that wait during a layer relax by amplitude damping, $\gamma = 1 - e^{-t/T_1}$, and phase damping. A gate lasts θ/Ω or θ/J, and large angles cost coherence.
- Readout error acts as a confusion matrix on the bit-string probabilities, $A = A_0 \otimes A_1 \otimes A_2$, $A_q = \begin{pmatrix} 1-\epsilon_{0\to1} & \epsilon_{1\to0} \\ \epsilon_{0\to1} & 1-\epsilon_{1\to0} \end{pmatrix}$.

A simulation applies $\hat{U}\hat{\rho}\hat{U}^\dagger$ and then the Kraus maps, at a memory cost of 4^N numbers.

A Noisy GHZ Circuit

The target $|\mathrm{GHZ}_N\rangle$ is prepared by $\hat{H}$ and a chain of $N-1$ CNOT gates. Its **fidelity** $F = \langle \mathrm{GHZ}_N | \hat{\rho} | \mathrm{GHZ}_N \rangle$ is the probability that the noisy register passes a test for the target. The model has $p_1 = 0.003$, $p_2 = 0.03$, relaxation of the waiting qubits in each layer and readout errors $\epsilon_{0\to1} = 0.01$, $\epsilon_{1\to0} = 0.03$.



For $N = 4$, $F = 0.915$, close to $(1-p_1)(1-p_2)^{N-1} = 0.910$, and $F = 0.884$ for $N = 5$. In this model each added CNOT lowers the fidelity. After readout a fraction 0.136 of the four-qubit strings differs from 0000 and 1111, and $\langle \hat{Z}_0 \hat{Z}_3 \rangle = 0.833$.

Coherent and Stochastic Errors

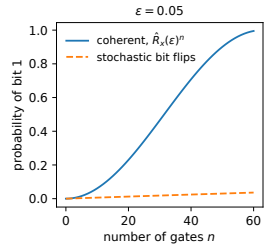
Two errors with equal infidelity per gate act k times on $|0\rangle$.

Coherent over-rotation $\hat{R}_x(\epsilon)$

Angles add, $\hat{R}_x(\epsilon)^k = \hat{R}_x(k\epsilon)$, and $1 - F = \sin^2 \frac{k\epsilon}{2} \approx k^2 \frac{\epsilon^2}{4}$. The infidelity grows as k^2 because amplitudes add.

Stochastic bit flip with $p = \sin^2 \frac{\epsilon}{2}$

An odd number of flips gives $1 - F = \frac{1}{2}\left[1 - (1-2p)^k\right] \approx k\,p$. Here probabilities add, and the infidelity grows as k.



At $\epsilon = 0.05$, $k = 20$ the infidelities are 0.23 and 0.012. Averaged over random compilations, randomised compiling turns coherent errors into stochastic Pauli errors.

J. J. Wallman and J. Emerson, Phys. Rev. A **94**, 052325 (2016), doi:10.1103/PhysRevA.94.052325

Fidelity and Trace Distance

Fidelity

For pure states $F = |\langle\varphi|\psi\rangle|^2$, the overlap probability.

For a pure target $|\psi\rangle$ and a prepared state $\hat{\sigma}$, possibly mixed,

$$F = \langle\psi|\hat{\sigma}|\psi\rangle.$$

Two mixed states need the Uhlmann fidelity, built from matrix square roots.

Trace distance

$$D(\hat{\rho}, \hat{\sigma}) = \tfrac{1}{2} \sum_k |\mu_k|,$$

where μ_k are the eigenvalues of the Hermitian matrix $\hat{\rho} - \hat{\sigma}$.

Given one copy of $\hat{\rho}$ or $\hat{\sigma}$ with equal probability, the best measurement identifies it with probability $\frac{1}{2}(1 + D)$. For one qubit, $D = \frac{1}{2}|\mathbf{r}_\rho - \mathbf{r}_\sigma|$.

A. Uhlmann, Rep. Math. Phys. **9**, 273 (1976), doi:10.1016/0034-4877(76)90060-4

Three Pairs of States

pair	fidelity F	trace distance D	best success $\frac{1}{2}(1+D)$
$\lvert+\rangle$ and the 50/50 mixture	0.50	0.50	0.75
Bell and the classical coin	0.50	0.50	0.75
Bell and dephased Bell, $p = 0.3$	0.58	0.42	0.71

- One copy of the Bell state or the coin is identified correctly with probability 3/4 at best, for example by measuring $\hat{X}\hat{X}$.
- After local dephasing the fidelity with $|\Phi^+\rangle$ is

$$F = \langle\Phi^+|\hat{\rho}|\Phi^+\rangle = \tfrac{1}{2}\big(1 + \underbrace{(1-2p)^2}_{\text{surviving coherence}}\big) = 0.58.$$

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Many Qubits



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Three-Qubit Gates and Universality

Toffoli gate as an evolution operator

The Toffoli gate flips qubit 2 when qubits 0 and 1 are both 1, $b_2 \to b_2 \oplus (b_0 \wedge b_1)$, the reversible AND of classical computing. Its generator is the projector on $|1\rangle|1\rangle|-\rangle$, and as for CNOT $e^{-i\pi\hat{P}} = \hat{I} - 2\hat{P}$, so

$$\widehat{\text{Toffoli}} = \exp\Big[- i\tfrac{\pi}{8}\,(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{X})\Big].$$

- Single-qubit rotations together with CNOT, or any other entangling two-qubit gate, build any N-qubit unitary, a property called **universality**.
- The *depth* of a circuit, the number of layers of gates acting in parallel, and its gate count measure its cost.
- Useful circuits are shallow, since each gate on a device adds noise.

A. Barenco *et al.*, Phys. Rev. A **52**, 3457 (1995), doi:10.1103/PhysRevA.52.3457

Summary

1. A qubit is a two-level system described by two complex amplitudes, and the Born rule turns them into the probabilities of a bit.
2. A gate is evolution under a chosen Hamiltonian for a chosen time. A single-qubit gate rotates the Bloch vector. A generator with $\hat{P}^2 = \hat{I}$ gives $e^{-ia\hat{P}} = \cos a\,\hat{I} - i \sin a\,\hat{P}$, and controlled gates are exponentials of projectors.
3. Other bases are measured by rotating first, and finite records give errors proportional to $1/\sqrt{M}$.
4. N qubits are 2^N amplitudes in binary order, qubit 0 most significant. Operators are sparse Kronecker products, and state-tensor contractions give the same results.

Summary, Continued

5. A pure two-qubit register with Schmidt rank 2 is entangled. The Bell state and the correlated coin give different records in the $\hat{X}\hat{X}$ and $\hat{Y}\hat{Y}$ settings.

6. The density matrix describes mixtures and parts of registers. Purity separates pure states from mixtures, and the reduced state measures the entanglement of a pure register.

7. Measurement has back-action and sends no signal. Errors are Kraus channels accumulating gate by gate. Fidelity and trace distance measure how far a state has moved.

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References



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Original Papers

- M. Born, Z. Phys. **37**, 863 (1926), doi:10.1007/BF01397477.
- E. Schmidt, Math. Ann. **63**, 433 (1907), doi:10.1007/BF01449770.
- I. I. Rabi, Phys. Rev. **51**, 652 (1937), doi:10.1103/PhysRev.51.652.
- G. Lüders, Ann. Phys. (Leipzig) **443**, 322 (1950), doi:10.1002/andp.19504430510.
- A. Uhlmann, Rep. Math. Phys. **9**, 273 (1976), doi:10.1016/0034-4877(76)90060-4.
- G. Lindblad, Commun. Math. Phys. **48**, 119 (1976), doi:10.1007/BF01608499.
- V. Gorini, A. Kossakowski and E. C. G. Sudarshan, J. Math. Phys. **17**, 821 (1976), doi:10.1063/1.522979.
- K. Kraus, *States, Effects, and Operations*, Lect. Notes Phys. **190** (Springer, 1983), doi:10.1007/3-540-12732-1.
- A. Barenco *et al.*, Phys. Rev. A **52**, 3457 (1995), doi:10.1103/PhysRevA.52.3457.
- J. M. Renes, R. Blume-Kohout, A. J. Scott and C. M. Caves, J. Math. Phys. **45**, 2171 (2004), doi:10.1063/1.1737053.

Original Papers on Qubit Couplings

- M. Saffman, T. G. Walker and K. Mølmer, Rev. Mod. Phys. **82**, 2313 (2010), doi:10.1103/RevModPhys.82.2313.
- D. Jaksch, J. I. Cirac, P. Zoller, S. L. Rolston, R. Côté and M. D. Lukin, Phys. Rev. Lett. **85**, 2208 (2000), doi:10.1103/PhysRevLett.85.2208.
- A. Sørensen and K. Mølmer, Phys. Rev. Lett. **82**, 1971 (1999), doi:10.1103/PhysRevLett.82.1971.
- K. Mølmer and A. Sørensen, Phys. Rev. Lett. **82**, 1835 (1999), doi:10.1103/PhysRevLett.82.1835.
- C. Rigetti and M. Devoret, Phys. Rev. B **81**, 134507 (2010), doi:10.1103/PhysRevB.81.134507.
- D. Loss and D. P. DiVincenzo, Phys. Rev. A **57**, 120 (1998), doi:10.1103/PhysRevA.57.120.

Labs and Reading

Labs

- Lab 064, one qubit and gates as rotations.
- Lab 065, the two-spin register and entanglement.
- Lab 066, density matrices, measurement and noise.
- Lab 067, from an atom in laser light to a qubit.
- Lab 068, from interacting qubits to two-qubit gates.

Textbooks and notes

- C. J. Foot, *Atomic Physics* (Oxford University Press, 2005).
- L. Allen and J. H. Eberly, *Optical Resonance and Two-Level Atoms* (Dover, 1987).
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2010), doi:10.1017/CBO9780511976667.
- J. Preskill, lecture notes on quantum computation, www.preskill.caltech.edu/ph229.